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List of Acronyms

31	AP	Average Precision
	BCE	Binary Cross-Entropy
	CNN	Convolutional Neural Network
	COCO	Common Objects in Context
	CoM	Center of Mass
	CSP	Cross Stage Partial
147	ECCV	European Conference on Computer Vision
76	FPGA	Field-Programmable Gate Array
	FPN	Feature Pyramid Network
	FPS	Frames Per Second
45	GAN	Generative Adversarial Network
	GCN	Graph Convolutional Network
	GDPR	General Data Protection Regulation
113	GIoU	Generalized Intersection over Union
	GPU	Graphics Processing Unit
	GRU	Gated Recurrent Unit
37	HOG	Histogram of Oriented Gradients
	INT8	8-bit Integer Quantization
	IoU	Intersection over Union
	IR	Infrared
	KL	Kullback-Leibler
49	LSTM	Long Short-Term Memory
	mAP	Mean Average Precision
	MLP	Multi-Layer Perceptron
	NETD	Noise Equivalent Temperature Difference
	NMS	Non-Maximum Suppression
	NPU	Neural Processing Unit
	OKS	Object Keypoint Similarity

	ONNX	Open Neural Network Exchange
	PAFPN	Path Aggregation Feature Pyramid Network
	QFL	Quality Focal Loss
	RGB	Red, Green, Blue
33	RLE	Residual Log-likelihood Estimation
	RNN	Recurrent Neural Network
	RoI	Region of Interest
	SimDR	Simple Disentangled Representation
33	SMPL	Skinned Multi-Person Linear Model
2	ST-GCN	Spatial-Temporal Graph Convolutional Network
	SVM	Support Vector Machine
151	ViT	Vision Transformer
	WHO	World Health Organization

Chapter 1 Introduction

1.1 Background and Motivation

Falls are a major health risk for older adults. The World Health Organization reports that falls are the second leading cause of accidental injury deaths worldwide, and people over 65 carry the highest burden [69]. Each year, about 37.3 million falls require medical attention, and 28%–35% of adults aged 65+ experience at least one fall annually (32%–42% for those over 70) [60, 35]. Critically, delayed assistance after a fall (a “long lie”) can cause severe secondary complications and markedly increased mortality risk [68, 16]. This motivates automated fall detection that is fast, reliable, and suitable for residential deployment.

Thermal infrared (IR) imaging has emerged as an exceptionally promising candidate that addresses this privacy–capability trade-off [63, 45]. Thermopile sensor arrays, such as the Heimann HTPA 60×40 module employed in the SensIR project [20], operate by measuring the far-infrared thermal radiation (wavelengths of approximately $8\text{--}14\text{ }\mu\text{m}$) emitted by objects in the scene. Because the human body maintains a core temperature significantly above the ambient temperature of typical indoor environments, individuals appear as distinctive warm regions—thermal silhouettes—against a cooler background in the sensor output. Crucially, the low spatial resolution of thermopile arrays (60×40 pixels) and the nature of far-infrared imaging together ensure that the captured data cannot reveal facial features, clothing patterns, skin color, or any other visually identifying information [39, 58]. This inherent privacy protection constitutes a decisive advantage over both RGB cameras and higher-resolution thermal cameras. Furthermore, thermal sensors are entirely independent of ambient illumination conditions, enabling consistent monitoring in complete darkness—a critical capability for nighttime fall detection in bedrooms and hallways where falls are particularly dangerous. The low data bandwidth associated with the 60×40 pixel resolution (each frame comprising only 2,400 temperature values) additionally reduces computational load, network transmission requirements, and storage overhead, rendering thermopile arrays well-suited for continuous long-term monitoring in embedded and edge computing environments [20].

The SensIR project, within which this thesis is situated, aims to develop a comprehensive deep learning framework for human pose estimation, activity recognition, and fall detection using a pair of Heimann HTPA 60×40 thermopile sensor arrays connected to an FPGA-accelerated edge computing device. The dual-sensor configuration extends the field of view, yielding a combined input resolution of approximately 114×40 pixels after spatial fusion. The system must operate at a minimum frame rate of 20 frames per second (fps) to enable real-time monitoring and prompt fall alerting, imposing stringent constraints on model complexity and computational efficiency. This application context motivates the central research question of this thesis: *How can we design a unified, lightweight deep learning framework that simultaneously performs person detection, body pose estimation, fall classification, and tracking from extremely low-resolution thermal imagery, while meeting the*

real-time throughput requirements of FPGA-based edge deployment?

1.2 Technical Challenges

Despite the inherent advantages of thermal infrared sensing for privacy-preserving indoor monitoring, the development of an accurate and efficient fall detection system operating on low-resolution thermopile data presents a constellation of interrelated technical challenges that distinguish this problem from conventional computer vision tasks.

Extreme spatial resolution constraints. The Heimann HTPA 60×40 sensor produces extremely low-resolution thermal images in which a human typically occupies only 10–30 pixels in height [45]. Fine-grained structural cues such as limb boundaries and joint articulations are heavily degraded, often collapsing into small amorphous blobs. Consequently, the backbone must extract semantically meaningful representations from severely information-limited inputs, with minimal spatial redundancy to support generalization [7].

Temporal ambiguity and inter-class confusion. Falls are brief events (0.5–2 s) that visually overlap with common daily activities such as sitting, lying down, or bending [15, 42]. Single-frame analysis is therefore insufficient. Reliable detection requires temporal modeling to capture characteristic motion dynamics (e.g., rapid descent followed by prolonged immobility), a challenge further amplified in the thermal domain due to the absence of texture and color cues.

Computational and latency constraints for edge deployment. Deployment on an FPGA-accelerated edge device imposes strict limits on model size, memory footprint, and arithmetic complexity [57, 7]. Achieving >20 fps requires inference latency below 50 ms per frame, while FPGA hardware offers limited on-chip memory, restricted operator support, and often fixed-point arithmetic [18]. Architectural design must therefore balance accuracy with computational efficiency and hardware compatibility.

Cascading errors in modular pipelines. Sequential pipelines (detection → pose → classification → tracking) suffer from error propagation [3]. Upstream inaccuracies degrade downstream performance, which is especially critical in fall detection. Joint optimization can mitigate this issue but introduces challenges in multi-task loss balancing and training stability.

1.3 Research Gap Analysis

A systematic examination of the existing literature on thermal infrared fall detection reveals several critical gaps that collectively motivate the present work. We identify five principal shortcomings in prior approaches.

Gap 1: Absence of keypoint-based structural reasoning. Most thermal fall detection systems rely on holistic image features or silhouette descriptors without modeling articulated body structure [45, 63]. Prior works employ handcrafted thermal features [39] or CNN classifiers on image patches [45], but none extract keypoint skeletons. In contrast, skeleton-based representations have shown strong discriminative power and view invariance in RGB

action recognition [72, 54]. Their application to low-resolution thermal imagery remains largely unexplored, despite offering a compact, interpretable, and privacy-preserving representation.

Gap 2: Limited temporal modeling. Many existing methods perform per-frame classification [63, 45] or rely on simple rule-based temporal heuristics [39]. Learned temporal models such as RNNs or temporal convolutional networks are rarely adopted. However, temporal modeling over skeleton sequences is crucial for distinguishing fall events from fall-like activities with similar static poses [14, 72]. The lack of learned temporal modeling limits robustness in practical scenarios.

Gap 3: No unified multi-task framework. To date, no prior work integrates detection, keypoint estimation, temporal classification, and multi-object tracking into a single jointly optimized architecture for thermal fall detection. Existing approaches use isolated or sequential modules [3], leading to redundant computation and error propagation. Although multi-task learning is known to improve representation sharing and generalization [4, 50], it has not been systematically applied in this domain.

Gap 4: Lack of hardware-aware design. Despite the edge-deployment motivation of low-resolution thermal sensing, few works explicitly account for embedded hardware constraints [57]. Architectures such as standard ResNets or VGG-style networks often exceed the computational budget of FPGA-based devices operating at real-time frame rates. A hardware–algorithm co-design strategy tailored to latency, memory, and arithmetic constraints is largely absent in current thermal fall detection literature.

1.4 Problem Formulation

We formulate fall detection as a composite perception problem over a temporal sequence of thermal infrared frames. Let $\mathbf{I}_t \in \mathbb{R}^{H \times W}$ denote the single-channel thermal image at discrete time step t , where $H = 40$ and $W = 114$ represent the fused resolution of the dual thermopile arrays. Each frame is resized to a canonical network input size of $H' \times W'$ (here 384×384) during both training and inference.

Given $\mathbf{I}_t$, the system predicts per-person outputs for all N_t individuals present in the scene. For each person $i \in \{1, \dots, N_t\}$, the model produces a bounding box $\mathbf{b}_t^{(i)} = (x^{(i)}, y^{(i)}, w^{(i)}, h^{(i)}) \in \mathbb{R}^4$ with confidence score $s_t^{(i)} \in [0, 1]$, and a set of $J = 7$ body keypoints

$$\mathbf{K}_t^{(i)} = \{(\hat{x}_j^{(i)}, \hat{y}_j^{(i)}, v_j^{(i)})\}_{j=1}^J, \quad (1.1)$$

where $(\hat{x}_j^{(i)}, \hat{y}_j^{(i)})$ are 2D keypoint coordinates and $v_j^{(i)} \in \{0, 1, 2\}$ indicates the visibility state (not labeled, occluded, or visible). In addition, the system outputs a fall probability $p_t^{(i)} \in [0, 1]$ for each person. To ensure temporal consistency, detections are associated across frames via a tracker that assigns a persistent identity $\text{id}_t^{(i)} \in \mathbb{Z}^+$.

Fall prediction is performed using a temporal window of normalized skeleton observations.

We define a bounding-box-normalized keypoint vector for person i at time t as

$$\mathbf{k}_t^{(i)} = \left[\frac{\hat{x}_1^{(i)} - x^{(i)}}{w^{(i)}}, \frac{\hat{y}_1^{(i)} - y^{(i)}}{h^{(i)}}, v_1^{(i)}, \dots, \frac{\hat{x}_J^{(i)} - x^{(i)}}{w^{(i)}}, \frac{\hat{y}_J^{(i)} - y^{(i)}}{h^{(i)}}, v_J^{(i)} \right] \in \mathbb{R}^{3J}, \quad (1.2)$$

which provides translation and scale invariance by normalizing coordinates relative to the enclosing bounding box. The temporal classifier operates on the sequence $\mathbf{k}_{t-T+1:t}^{(i)} = [\mathbf{k}_{t-T+1}^{(i)}, \dots, \mathbf{k}_t^{(i)}] \in \mathbb{R}^{T \times 3J}$ to estimate the fall probability

$$p_{\text{fall}} = \sigma(\mathbf{w}_c^T \mathbf{o} + b_c). \quad (1.3)$$

where f_{action} denotes a GRU-based temporal model with parameters θ_{action} , and $\sigma(\cdot)$ is the sigmoid activation function. The tracking identity $\text{id}_t^{(i)}$ is used to maintain the correct per-person temporal buffer across frames.

The overall training objective jointly optimizes detection, pose estimation, and temporal fall classification in a unified multi-task setting:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}}(\theta_{\text{det}}) + \mathcal{L}_{\text{pose}}(\theta_{\text{pose}}) + \mathcal{L}_{\text{action}}(\theta_{\text{action}}), \quad (1.4)$$

where θ_{det} , θ_{pose} , and θ_{action} are learnable parameters for detection, pose estimation, and action classification, respectively, with feature sharing through a common backbone/neck. The individual loss terms and training details are described in Chapter Method.

1.5 Rationale for Unified Multi-Task Learning

Our system is organized as an end-to-end trainable *multi-head* network for person detection, pose estimation, and fall classification, with object tracking applied as an inference-time association module. Concretely, the detector, keypoint head, and action head share a common backbone/neck and are optimized jointly with a combined objective, while tracking is performed after the network forward pass to maintain identity consistency over time. Under this setting, multi-task learning is adopted primarily for efficiency and representation sharing, rather than claiming a fully coupled “all-in-one” solution that learns tracking jointly.

First, sharing a single feature backbone amortizes the cost of feature extraction across detection, pose estimation, and action classification. In contrast to modular designs where each stage may maintain its own feature extractor, our shared backbone computes multi-scale features once per frame and feeds them to the detection and pose branches, while the action head leverages the resulting pose outputs. This reduces redundant computation and memory traffic, which is important for real-time inference under edge constraints [4].

Second, joint optimization of detection, pose, and action provides a practical regularization effect and can improve generalization on modest-sized datasets [50]. Compared with large-scale benchmarks such as COCO [30] or ImageNet [11], the available thermal training

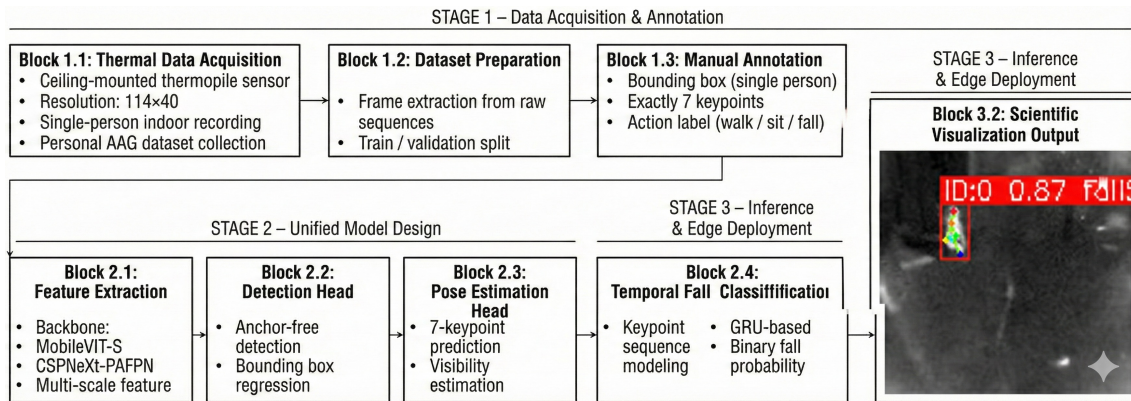


Figure 1.1: Overall research framework of the proposed thermal infrared fall detection system. The pipeline consists of three major stages: (1) thermal data acquisition and annotation of a single-person dataset using a 114x40 thermopile sensor; (2) unified multi-task model design integrating detection, seven-keypoint pose estimation, and temporal fall classification; and (3) real-time inference and edge deployment with fall probability output.

data are relatively limited. By learning shared features that must support both localization (detection), structural understanding (pose), and temporal decision making (action), the network is encouraged to avoid overly task-specific shortcuts. In particular, the pose estimation objective encourages the backbone to preserve body-structure cues that can also benefit fall-related classification.

Third, although our tracking module is not part of end-to-end training, joint training of detection and pose can still reduce error amplification within the perception front-end. More accurate person localization improves the RoI features used by the keypoint head, and better keypoint quality provides a cleaner input to the temporal action head. In this sense, the action loss can provide an additional learning signal that complements the pose loss, while we avoid claiming full cross-module optimality across tracking, which remains a post-processing component.

Finally, this design simplifies deployment. Instead of maintaining separate detection and pose networks on the edge device, the main perception components are consolidated into a single model that can be exported and optimized as a unit (e.g., quantization and hardware mapping), while tracking remains a lightweight inference-time module. This reduces engineering overhead and potential synchronization issues compared with fully modular pipelines.

1.6 Approach Overview

Building upon the analysis of existing gaps and the articulated design principles, we propose **RTMDetWithPose**, a unified end-to-end deep learning framework that simultaneously addresses human body detection, seven-keypoint pose estimation, binary fall classification, and multi-object tracking from thermal infrared images captured by ther-

20 mobile sensor arrays. The framework is designed from the outset to satisfy the competing demands of high perceptual accuracy and low computational cost, making it suitable for real-time deployment on FPGA-accelerated edge devices.

2 The architectural design of RTMDetWithPose is organized around four tightly integrated components that share a common feature extraction pathway. The feature extraction pathway consists of a **MobileViT-S backbone** [41], which combines the local receptive fields of depthwise separable convolutions with the global context modeling of vision transformers in a parameter-efficient hybrid architecture, and a **CSPNeXt-PAFPN neck** [38, 31] that aggregates features from multiple backbone stages into three output scales suitable for detecting persons at varying distances from the sensor.

4 11 An anchor-free **RTMDet detection head** [38] operates on the multi-scale feature maps to predict person bounding boxes and confidence scores, using Quality Focal Loss [27] for classification and GIoU Loss [48] for box regression with a dynamic soft label assignment strategy. For each detected person, a **RoIAlign module** [19] extracts a fixed-size feature patch from the shared multi-scale features, which is then processed by the pose estimation head.

28 123 For pose estimation, we conduct a systematic investigation of five alternative head designs spanning the three major paradigms in keypoint localization: heatmap-based methods [43, 70], integral regression [56, 44], and Disentangled Representation [28]. Our final architecture adopts the **SimDR formulation** [28], which decomposes the 2D keypoint localization problem into two independent 1D classification tasks along the horizontal and vertical axes. This approach eliminates the quadratic memory cost of 2D heatmap representations, avoids the quantization errors inherent in discrete argmax decoding, and achieves state-of-the-art accuracy with substantially fewer operations—properties that are particularly advantageous in the low-resolution, resource-constrained thermal domain.

A lightweight **GRU-based temporal action classifier** [9] consumes the sequence of bounding-box-normalized keypoint coordinates for each tracked individual, enabling the system to model the temporal dynamics of body motion and distinguish genuine falls from visually similar non-fall activities. The temporal sequences are maintained by a **ByteTrack multi-object tracker** [74] that uses Kalman-filter-based motion prediction and a two-stage association strategy to preserve identity consistency even through temporary occlusions and low-confidence detections.

56 1.7 Contributions

The principal contributions of this thesis are as follows:

- 4 4 3 1. **Unified end-to-end multi-task framework.** We present RTMDetWithPose, a single-model architecture that jointly performs object detection, seven-keypoint pose estimation, skeleton-based fall classification, and multi-object tracking from low-resolution thermal infrared imagery. To the best of our knowledge, this is the first framework to integrate all four capabilities within a unified, jointly trained architecture specifically designed for the thermopile sensor domain. The unified

design enables shared feature computation, beneficial gradient interactions between tasks, and a compact deployment footprint suitable for edge devices.

2. **Systematic pose estimation head comparison.** We conduct a rigorous ablation study comparing five distinct pose estimation head architectures—2D Gaussian heatmaps, foreground-weighted heatmaps with visibility prediction, soft-argmax integral regression, SimDR-based 1D coordinate classification, and direct coordinate regression—within the unified framework. This comparison provides the first systematic evaluation of modern pose estimation paradigms in the low-resolution thermal domain, and demonstrates that the SimDR formulation achieves the most favorable accuracy–efficiency trade-off under the extreme resolution constraints of thermopile sensor data.
3. **Lightweight temporal fall classification with skeleton-based GRU.** We introduce a computationally efficient GRU-based temporal action classifier that operates on bounding-box-normalized skeleton sequences, with per-track keypoint buffering maintained by an integrated ByteTrack multi-object tracker. This design enables temporally coherent fall detection in multi-person scenarios while adding negligible computational overhead to the overall inference pipeline. The normalization scheme ensures translation and scale invariance of the skeleton features, and the GRU architecture is selected specifically for its suitability to the minimal skeleton dimensionality ($J = 7$ keypoints) of the thermopile domain.
4. **Validation on the SensIR thermal infrared dataset.** We validate the complete framework on the SensIR dataset, a thermal infrared dataset captured with Heimann HTPA 60×40 thermopile sensor arrays annotated with bounding boxes, seven-keypoint poses, and fall/non-fall labels. Our experiments demonstrate that the proposed system achieves competitive performance in detection, pose estimation, and fall classification while maintaining a model complexity compatible with real-time inference on edge hardware operating at >20 fps.

1.8 Thesis Organization

The remainder of this thesis is organized as follows. Chapter Related work provides a comprehensive review of related work, covering five domains: anchor-free object detection, human pose estimation methods, skeleton-based action recognition, multi-object tracking algorithms, and fall detection systems with emphasis on thermal infrared approaches. Chapter Method presents the proposed RTMDetWithPose framework in detail, describing the backbone, neck, detection head, RoI extraction, pose estimation head variants, temporal action classifier, tracking integration, and multi-task training strategy. Chapter Experiments reports experimental results on the SensIR dataset, including implementation details, ablation studies on pose estimation head design, detection and pose estimation metrics, fall classification performance, and tracking analysis. Finally, Chapter Conclusion summarizes the findings, discusses limitations of the current system, and outlines directions for future work including fine-grained multi-class action recognition, graph-based temporal modeling, and model quantization for FPGA deployment.

Chapter 2 Relative Work

This chapter surveys the four main research areas that collectively form the foundation of the proposed fall detection framework: thermal infrared human detection (Section 2.1), human pose estimation (Section 2.2), skeleton-based action recognition (Section 2.3), and vision-based fall detection systems (Section 2.4). For each area, this chapter traces the evolution from classical approaches to modern deep learning methods, identifies remaining challenges, and positions the contributions of this thesis within the broader literature.

2.1 Thermal Infrared Human Detection

Advantages of Thermal Imaging

Thermal infrared imaging has gained substantial attention for human-centric perception tasks in recent years, driven by three fundamental advantages over conventional visible-spectrum cameras. First, thermal sensors capture radiated body heat rather than reflected light, which renders the resulting imagery inherently privacy-preserving: facial features, clothing textures, and other personally identifiable visual cues are largely suppressed, making it practically impossible to identify individuals from the thermal signature alone [17]. This property is particularly important in the context of indoor elderly monitoring, where compliance with data protection regulations such as the European General Data Protection Regulation (GDPR) imposes strict constraints on the collection and processing of biometric data. Second, because thermal emission is independent of ambient illumination, detection performance remains stable across complete darkness, strong backlighting, and varying indoor lighting conditions—a critical requirement for systems that must operate around the clock without interruption. Third, the high thermal contrast between the human body (core temperature approximately 37°C) and typical indoor backgrounds (20–25°C) provides a strong detection cue even at very low spatial resolutions. This contrast is the fundamental physical principle exploited by thermopile sensor arrays, such as the Heimann HTPA 60×40 [20], which produce ultra-low-resolution thermal images that maximize privacy while retaining sufficient information for human body analysis.

Gade and Moeslund [17] provided a comprehensive survey of thermal camera applications across domains including surveillance, medical imaging, and industrial inspection, noting that the unique characteristics of thermal imagery—particularly the absence of texture and color—require specialized processing pipelines distinct from those developed for visible-spectrum data. Early studies further demonstrated that meaningful behavioral information can be extracted from low-resolution infrared sensors even when spatial resolution is insufficient to reveal personal identity.

Classical and Deep Learning Detection Methods

Early thermal human detection methods adopted classical machine learning pipelines combining handcrafted feature descriptors with discriminative classifiers. The Histogram of Oriented Gradients (HOG) descriptor introduced by Dalal and Triggs [10] became a foundational representation for pedestrian detection, capturing local gradient structure in a way that generalizes across appearance variation. When paired with a Support Vector Machine (SVM) classifier, HOG-based detectors achieved reasonable performance on thermal imagery but remained limited in their ability to handle diverse body poses, partial occlusions, and the scale variations inherent in practical monitoring scenarios.

The release of large-scale multispectral benchmarks substantially accelerated research in thermal detection. The KAIST Multispectral Pedestrian Detection dataset [21] provided precisely aligned RGB–thermal image pairs captured across diverse urban scenes during both daytime and nighttime conditions, enabling systematic study of cross-modal fusion strategies and demonstrating that thermal imagery dramatically improves detection robustness under poor illumination. The LLVIP dataset [23] further extended this line of work by offering high-quality paired visible–infrared images specifically targeting low-light scenarios, where the performance gap between thermal and visible modalities is most pronounced. These benchmarks collectively established thermal human detection as a mature research direction with well-characterized advantages over single-modality visible-spectrum approaches.

With the advent of deep learning, convolutional neural network (CNN)-based object detectors rapidly surpassed classical methods. Two-stage detectors such as Faster R-CNN [47], which decouples region proposal generation from classification, demonstrated strong accuracy on thermal inputs but introduced computational overhead that limits real-time deployment. Single-stage detectors including YOLO [46] and SSD [32] eliminated the proposal stage entirely, directly predicting bounding boxes and class scores from dense feature maps, achieving substantial improvements in inference speed at a modest accuracy cost. Devaguptapu *et al.* [12] specifically addressed the challenge of limited thermal training data by proposing a pseudo multi-modal approach that transfers representations learned from large-scale visible-spectrum datasets to thermal detectors, demonstrating that cross-domain knowledge transfer can effectively bootstrap thermal detection performance.

Real-Time Detection and the Proposed Approach

Among recent architectures designed explicitly for real-time applications, RTMDet [38] achieves an excellent speed–accuracy trade-off through several design innovations. It employs a CSPNeXt-based backbone with a Path Aggregation Feature Pyramid Network (PAFPN) [31] neck that enhances multi-scale feature fusion through bidirectional information flow. The detection head uses an anchor-free formulation that eliminates the need for pre-defined anchor boxes, simplifying the pipeline and improving generalization to objects of varying aspect ratios—a property particularly beneficial for detecting humans in the diverse poses encountered during fall events (e.g., standing, sitting, lying down). Training is supervised by Quality Focal Loss [27], which jointly models classification confidence and localization quality, and GIoU loss [48] for bounding box regression.

In the proposed framework, RTMDet is adopted as the detection architecture, but with the original CSPNeXt backbone replaced by MobileViT-S [41], a lightweight hybrid architecture that interleaves depthwise separable convolutions (inherited from MobileNetV2 [51]) with vision transformer blocks. MobileViT-S combines the local feature extraction efficiency of convolutional operations with the global contextual modeling capacity of self-attention, achieving a favorable accuracy–latency trade-off on mobile and edge hardware. The deployment target of the system—an FPGA-accelerated edge device running at >20 FPS—necessitates this lightweight backbone choice, as efficient neural network deployment on resource-constrained hardware remains a significant engineering challenge [57].

2.2 Human Pose Estimation

Top-Down versus Bottom-Up Paradigms

Human pose estimation aims to localize anatomical keypoints of the human body from images and constitutes a fundamental building block for higher-level tasks such as action recognition, person re-identification, and human-computer interaction. Existing methods are broadly classified into two paradigms. Top-down methods [70, 55] first detect individual persons via a bounding box detector and subsequently estimate keypoints within each cropped and resized region of interest (RoI). Bottom-up approaches [3] detect all keypoints in the image simultaneously and group them into individual skeletons through associative embedding or part affinity fields. Cao *et al.* [3] demonstrated that bottom-up methods can achieve real-time multi-person pose estimation by predicting part affinity fields that encode limb associations, but their accuracy on individual instances tends to be lower than that of top-down methods, particularly when persons are at different scales or are partially occluded.

Top-down methods generally achieve higher per-instance accuracy and integrate naturally with detection-based pipelines, as the detector provides a well-defined spatial context (the bounding box) within which keypoint estimation operates. The proposed framework adopts the top-down paradigm, applying keypoint estimation to RoI-pooled features [19] extracted from each detected bounding box.

Heatmap-Based Methods

Within the top-down framework, heatmap-based methods have long dominated the field. The core idea is to predict, for each keypoint, a 2D spatial probability map (heatmap) in which the peak value indicates the most likely keypoint location. The Stacked Hourglass Network [43] pioneered this approach by introducing repeated bottom-up and top-down processing modules with intermediate supervision, enabling the network to refine predictions progressively at multiple scales. The architecture's symmetrical encoder-decoder design with skip connections became a template for subsequent work.

SimpleBaseline [70] demonstrated a surprising result: a straightforward architecture comprising a ResNet backbone followed by just three deconvolutional layers could achieve accuracy competitive with far more complex designs, suggesting that the heatmap output

representation itself—rather than architectural novelty—is the key driver of performance. This finding underscored the effectiveness and robustness of the Gaussian heatmap formulation as a supervision target.

HRNet (High-Resolution Network) [55] further advanced the state of the art by fundamentally rethinking the backbone architecture. Rather than encoding the input to a low-resolution representation and then decoding back to high resolution (as in hourglass-style networks), HRNet maintains high-resolution feature representations in parallel throughout the network, with repeated multi-scale feature fusion. This design avoids the information loss inherent in spatial downsampling and upsampling, yielding more spatially precise heatmap predictions. HRNet achieved state-of-the-art results on the COCO keypoint benchmark [30] and became one of the most widely used backbones for pose estimation.

More recently, ViTPose [71] demonstrated that plain vision transformers, pre-trained on large-scale image classification datasets, can serve as highly effective backbones for pose estimation without any pose-specific architectural modifications. The global receptive field of self-attention enables ViTPose to capture long-range spatial dependencies between distant body parts, which is particularly beneficial for estimating the positions of limbs relative to the torso.

Limitations of 2D Heatmaps and Coordinate Refinement

Despite the widespread success of heatmap-based methods, 2D heatmap representations suffer from an inherent quantization error: the spatial resolution of the output heatmap is typically $4\times$ to $8\times$ lower than the input image, and the standard argmax decoding step discretizes predicted coordinates to integer grid locations within this coarse spatial grid. For high-resolution inputs, this quantization may be acceptable, but for the ultra-low-resolution thermal images targeted in this thesis (114×40 pixels), even small quantization errors can represent a significant fraction of the person's spatial extent.

Several approaches have been proposed to mitigate this limitation. DARK Pose (Distribution-Aware Coordinate Representation) [73] introduced a principled sub-pixel refinement technique based on Taylor expansion of the log-likelihood around the heatmap peak, achieving measurable improvements in localization accuracy without modifying the network architecture. The integral regression (soft-argmax) approach, independently proposed by Sun *et al.* [56] and Nibali *et al.* [44], replaced the non-differentiable argmax operation with a differentiable spatial expectation computed over the softmax-normalized heatmap. This formulation enables end-to-end gradient flow from coordinate-level supervision through the heatmap representation, combining the spatial structure of heatmaps with the precision of direct coordinate regression.

Direct Regression and SimDR

At the opposite end of the design spectrum, direct coordinate regression methods predict keypoint positions directly from feature vectors without any intermediate spatial representation. DeepPose [61], the pioneering work in this category, employed a cascade of

51 CNN regressors to iteratively refine keypoint predictions. While conceptually simple and computationally efficient, regression-based methods have historically underperformed heatmap approaches due to the difficulty of learning the highly nonlinear mapping from high-dimensional appearance features to low-dimensional coordinate outputs. Li *et al.* [26] proposed Residual Log-likelihood Estimation (RLE) to address the fundamental ill-posedness of direct regression by modeling the output distribution rather than predicting point estimates, narrowing the performance gap with heatmap methods.

A recent breakthrough that elegantly bridges the gap between heatmaps and regression is SimDR (Simple Disentangled Representation), presented by Li *et al.* [28] as an oral paper at ECCV 2021. SimDR poses the provocative question: *is a 2D heatmap representation even necessary for accurate keypoint localization?* Their answer reformulates pose estimation as two independent 1D classification problems—predicting the horizontal bin index and the vertical bin index for each keypoint—thereby replacing the quadratic-cost 2D heatmap ($H \times W$ per keypoint) with two linear-cost 1D probability vectors ($H + W$ per keypoint). By using a configurable bin resolution (scale factor s) that exceeds the spatial resolution of the feature map, SimDR achieves sub-pixel accuracy while entirely avoiding 2D spatial quantization error. The 1D classification targets are soft Gaussian distributions supervised by KL divergence loss, providing a smoother optimization landscape than the MSE loss typically used for 2D heatmaps.

In the proposed framework, the SimDR formulation is applied to 48×48 RoI features with scale factor $s = 2$ (yielding 96 bins per axis), enabling sub-pixel keypoint localization within the extremely limited spatial resolution of the thermal input images. An auxiliary visibility branch predicts per-keypoint occlusion flags via binary cross-entropy loss, following the three-level visibility weighting scheme of CrowdPose [25] (invisible $\rightarrow$ ignored; occluded $\rightarrow$ reduced weight; visible $\rightarrow$ full weight). Through systematic ablation experiments (Chapter Experiments), this approach is compared against four alternative head designs—standard 2D Gaussian heatmap regression, foreground-weighted heatmap with visibility prediction, soft-argmax integral regression, and direct coordinate regression—demonstrating that SimDR achieves the best accuracy–efficiency trade-off for the low-resolution thermal input scenario.

146 2.3 Skeleton-Based Action Recognition

Motivation and Advantages

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92
90 Skeleton-based action recognition has emerged as a prominent research direction that leverages the compact, structured representation of human body keypoints for temporal activity understanding. Compared to appearance-based methods such as two-stream networks and 3D convolutional networks that operate on raw pixel intensities or optical flow, skeleton representations offer several compelling advantages. They are inherently robust to background clutter, illumination changes, and appearance variation, since only the geometric configuration of body joints is retained. They are also highly compact—a skeleton with K keypoints requires only $2K$ or $3K$ values per frame (2D or 3D coordinates), compared to millions of pixel values in a video frame—making skeleton-based methods

computationally efficient and well suited for edge deployment. Furthermore, in privacy-sensitive thermal infrared applications where texture and color information are absent, the skeleton representation naturally aligns with the available information: the thermal image primarily reveals the spatial extent and configuration of the body, which is precisely what keypoints encode.

RNN and LSTM Approaches

Early deep learning approaches to skeleton-based action recognition employed recurrent neural networks (RNNs) to model the temporal evolution of keypoint sequences. Du *et al.* [14] proposed a hierarchical RNN architecture that decomposes the skeleton into five anatomical sub-groups (two arms, two legs, and trunk), processes each sub-group with a separate RNN branch, and hierarchically fuses the temporal features in a tree structure. This decomposition exploits the natural hierarchy of the human body and demonstrated the viability of sequential modeling for activity classification on early skeleton benchmarks. The Gated Recurrent Unit (GRU) [9] and Long Short-Term Memory (LSTM) networks were subsequently applied across various skeleton benchmarks, offering mechanisms to capture long-range temporal dependencies through learned gating functions that control information flow. These recurrent approaches established strong baselines, particularly for datasets with relatively few keypoints and simple skeleton topologies.

Graph Convolutional Networks

The introduction of Spatial-Temporal Graph Convolutional Networks (ST-GCN) by Yan *et al.* [72] marked a paradigm shift in the field. ST-GCN formulates the skeleton as a spatiotemporal graph in which joints are nodes and bones (anatomical connections) are edges, and applies graph convolutions that simultaneously capture spatial dependencies among joints within a single frame and temporal dynamics across consecutive frames. By operating directly on the graph structure, ST-GCN avoids the need to flatten the skeleton into a 1D vector (as required by RNNs) and can naturally model the relational structure between body parts. ST-GCN achieved state-of-the-art results on the NTU RGB+D dataset [53], one of the largest benchmarks for skeleton-based action recognition, and on the Kinetics skeleton dataset.

Subsequent works extended the ST-GCN paradigm in several important directions. Two-Stream Adaptive Graph Convolutional Networks (2s-AGCN) [54] introduced two key innovations: (1) learnable adjacency matrices that adapt the graph topology during training, allowing the network to discover task-relevant joint relationships beyond the predefined anatomical connections; and (2) a two-stream architecture that processes both joint position data and bone vector data (computed as difference vectors between connected joints) in parallel, fusing the two streams for final classification. The bone stream provides complementary information about limb orientations and lengths that is not directly encoded in isolated joint positions. MS-G3D [33] further improved multi-scale temporal modeling through disentangled multi-scale graph convolutions that capture temporal dependencies at different time scales simultaneously, addressing the limitation of ST-GCN's fixed temporal receptive field. Channel-wise Topology Refinement Graph

Convolution (CTR-GC) [8] refined the graph topology at the channel level, learning different relational structures for different feature channels and achieving strong performance on NTU RGB+D benchmarks.

Input Representations and Multi-Stream Fusion

Beyond architectural innovations, the choice of input representation has proven critical for skeleton-based recognition performance. Raw joint coordinates provide the basic positional information but are sensitive to the person's absolute position and scale within the frame. Bone features, computed as the difference vectors between connected joint pairs, encode limb orientation and length information that is invariant to translation. Joint velocity features, obtained by computing first-order temporal differences of joint positions between consecutive frames, capture motion dynamics and are particularly informative for distinguishing actions that share similar static poses but differ in movement patterns (e.g., standing still vs. standing up). Bone velocity features combine both geometric and temporal information. Multi-stream fusion of these four input modalities—joint positions, bone vectors, joint velocities, and bone velocities—has become standard practice in high-performing systems [54, 8], with each stream processed by a separate network and the outputs fused (typically by score averaging) for final prediction.

12 The NTU RGB+D dataset [53], captured using Microsoft Kinect depth sensors with 25 body keypoints per person, has served as the primary benchmark for evaluating skeleton-based methods. However, the 25-keypoint Kinect skeleton is substantially richer than the 7-keypoint skeleton available in the thermal infrared setup of this work, and reduced skeleton complexity presents unique challenges for recognition algorithms that rely on rich inter-joint spatial structure.

The Proposed Approach

4 The proposed framework adopts a lightweight GRU-based temporal classifier rather than a full graph convolutional network. This design choice is motivated by the observation that the skeleton in this work consists of only seven keypoints (head, left/right shoulders, left/right hands, hip, and lower body reference) with a simple, largely linear topology, for which the expressive power of graph convolutions offers diminishing returns relative to their computational overhead. For a 7-joint skeleton, the spatial graph has very limited structure—far less than the 25-joint Kinect skeleton for which ST-GCN was designed—and a GRU operating on the flattened joint feature vector can effectively capture the relevant temporal patterns. The feature representation evolves through several design iterations during development: from raw normalized keypoint coordinates, to velocity-augmented features that capture inter-frame motion dynamics, to bone-angle features that encode geometric relationships between limbs, and finally to a variant that fuses skeleton features with appearance-level RoI embeddings extracted from the detection backbone. This multi-task learning approach [4] allows the action classifier to benefit from shared feature representations learned jointly with the detection and pose estimation tasks.

2.4 Fall Detection Systems

Sensing Modalities

Fall detection has been the subject of extensive research over the past two decades, driven by the significant health consequences of undetected falls among the elderly [69, 60]. Falls are a leading cause of injury-related morbidity and mortality among older adults, and the duration of time spent on the ground after a fall—the so-called “long lie”—is strongly correlated with adverse outcomes including hypothermia, dehydration, and death [35]. With the global population aging rapidly, the demand for reliable, automated fall detection systems in indoor environments has become increasingly urgent. Comprehensive surveys of fall detection technologies have been provided by Mubashir *et al.* [42], Wang *et al.* [66], and Igual *et al.* [22], each offering different taxonomies and evaluation perspectives.

Existing fall detection systems can be categorized by their sensing modality into three broad classes: wearable sensors, ambient sensors, and vision-based systems. Wearable sensor-based approaches employ tri-axial accelerometers, gyroscopes, or inertial measurement units (IMUs) attached to the body (typically at the waist, wrist, or chest) to detect falls by analyzing abrupt changes in acceleration magnitude, angular velocity, and body orientation [42]. While these methods can achieve high sensitivity and specificity under controlled conditions, they fundamentally depend on user compliance: the individual must consistently wear, charge, and maintain the device. Studies have reported low long-term adherence rates among elderly populations, particularly those with cognitive impairments, severely limiting the practical utility of wearable solutions in unsupervised real-world settings [35].

Ambient sensor-based approaches offer a non-intrusive alternative by instrumenting the environment rather than the person. Floor-mounted pressure or vibration sensors can detect the impact of a fall event, while acoustic sensors can capture characteristic sound patterns associated with falls. However, these methods are limited in spatial coverage, lack contextual information about the nature of the event, and are prone to false alarms in realistic home environments.

Vision-Based Fall Detection

Vision-based fall detection has attracted growing interest due to its ability to monitor large areas continuously without requiring user cooperation or environmental instrumentation. Early approaches using RGB cameras relied on handcrafted features extracted from video sequences. Rougier *et al.* [49] proposed a fall detection method based on tracking changes in the human silhouette shape over time, using Gaussian mixture models for foreground segmentation and shape matching for event classification. Charfi *et al.* [5] systematically evaluated spatio-temporal descriptors for real-time fall detection, comparing SVM and AdaBoost classifiers on features including motion history images, aspect ratio trajectories, and velocity profiles. These handcrafted approaches established important baselines but were limited by their reliance on manually designed features that may not generalize across diverse environments and fall patterns.

The application of deep learning to vision-based fall detection brought substantial performance improvements, with CNNs learning discriminative spatio-temporal representations directly from video data. Depth cameras (e.g., Microsoft Kinect) provided an intermediate solution that captures 3D body structure while partially preserving privacy through the absence of color and texture information [40]. However, depth sensors have limited operational range (typically 0.5–4.5 m for Kinect), are susceptible to interference from reflective and transparent surfaces, and still capture sufficient geometric detail to potentially identify individuals through body shape analysis.

Thermal Infrared Fall Detection

Thermal infrared cameras have emerged as the most compelling modality for fall detection in privacy-critical environments, combining the spatial awareness of vision-based systems with the anonymity guarantees required by data protection regulations. Several research groups have demonstrated the viability of thermal fall detection at various sensor resolutions.

At the higher end of the resolution spectrum, Vadivelu *et al.* [63] explored thermal imaging for elderly fall detection using standard thermal cameras, demonstrating robust performance across varying lighting conditions. Tao *et al.* [58] proposed a deep learning pipeline for real-time human body segmentation and fall detection from thermal infrared images, combining semantic segmentation with temporal analysis of body orientation changes. Nogas *et al.* [45] demonstrated that CNNs trained on thermal images from ceiling-mounted cameras can reliably detect falls even at relatively low resolutions, achieving strong performance with compact network architectures suitable for embedded deployment.

At the ultra-low-resolution end—which offers the strongest privacy guarantees—Trofimova *et al.* [62] investigated activity recognition using low-resolution infrared array sensors, showing that basic activities including sitting, standing, lying, and walking can be distinguished even from coarse thermal signatures. Tateno *et al.* [59] specifically addressed privacy-preserving fall detection in smart homes using low-resolution thermal array sensors, proposing a CNN-based classification approach that achieves reliable fall detection while ensuring that the spatial resolution is too low to reveal any personally identifiable information. The present work builds upon this body of research by operating on 114×40 pixel thermal images produced by dual Heimann HTPA sensors [20], a resolution that provides sufficient spatial detail for 7-keypoint pose estimation while remaining firmly within the privacy-preserving regime.

Binary versus Multi-Class Formulation

A fundamental design question in fall detection is the choice of problem formulation: binary classification (fall vs. non-fall) versus multi-class action recognition from which falls are inferred indirectly [66]. The binary formulation directly optimizes for the target event by training a classifier specifically to distinguish falls from all other activities. This approach avoids error propagation from intermediate action labels and can concentrate model capacity on the decision boundary that matters most, but it may miss contextual

information about the transition dynamics leading up to a fall (e.g., the distinction between falling while walking versus falling while trying to stand). Multi-class approaches [75] can capture finer activity semantics and provide richer behavioral analytics, but they require more extensive annotation, increase the number of output classes over which the classifier must discriminate, and risk diluting the classifier's sensitivity to the relatively rare fall class—a concern exacerbated by the severe class imbalance inherent in realistic fall datasets, where falls constitute only a small fraction of all observed activities.

Rule-based post-processing heuristics have been employed in both paradigms to reduce false positive rates. Common heuristics include detecting rapid downward displacement of the head keypoint (a kinematic signature of falling), sustained horizontal body orientation following a motion event (distinguishing falls from intentionally lying down), prolonged stillness in a lying-down posture (detecting “long lie” events), and sudden changes in bounding box aspect ratio from vertical to horizontal [5, 49]. These rule-based methods complement learned classifiers by encoding domain knowledge about the biomechanics and temporal dynamics of fall events.

The Proposed Approach

In the proposed system, fall detection is formulated as a binary temporal classification task, directly predicting a fall probability from a sequence of normalized skeleton keypoints using a GRU-based classifier. Through ablation experiments in Chapter Experiments, it is demonstrated that this direct binary formulation outperforms indirect inference from the 10-class activity taxonomy available in the SensIR dataset when evaluated on fall-specific metrics (precision, recall, and F1-score). To address the inherent class imbalance—fall events are substantially rarer than normal activities in any realistic monitoring scenario—a weighted binary cross-entropy loss with positive class weighting ($w_+ = 5.0$) is employed, following standard practices for imbalanced classification in medical and safety-critical domains.

The temporal integration facilitated by ByteTrack [74] multi-object tracking, built upon the Kalman Filter motion model of SORT [2], enables the classifier to operate over variable-length keypoint sequences (1–30 frames). This temporal context allows the GRU to observe the dynamic transition pattern—the rapid downward displacement of the head, the rotation of the body axis from vertical to horizontal, the deceleration following impact—that distinguishes genuine falls from similar static configurations such as a person lying in bed. The end-to-end integration of detection, pose estimation, action classification, and tracking within a single jointly trained framework is a distinguishing feature of the proposed approach, enabling shared feature representations and efficient inference on the resource-constrained edge hardware targeted by the SensIR project.

Chapter 3 Method

3.1 Overall Architecture

The proposed system, which we refer to as **RTMDetWithPose**, is a unified multi-task framework that jointly performs human detection, seven-keypoint pose estimation, temporal fall classification, and multi-object tracking within a single architecture. The framework is organized into three functional stages:

Stage 1 (Detection + Pose Estimation): A single-frame thermal IR image, resized to 384×384 pixels, is processed by a shared MobileViT-S backbone and CSPNeXtPAFPN neck to produce multi-scale feature maps at strides $\{8, 16, 32\}$. An anchor-free RTMDet detection head localizes human subjects, producing bounding boxes and confidence scores. For each detected person, RoI-aligned features of size 48×48 are extracted from the multi-scale feature pyramid and fed to a SimDR-based pose estimation head, which predicts seven keypoint coordinates along with per-keypoint visibility scores.

Stage 2 (Temporal Action Classification): For each tracked person, the estimated keypoint coordinates are normalized relative to the bounding box and augmented with velocity features. The resulting temporal sequence is processed by a dual-task GRU classifier that simultaneously predicts 4-class activity and binary fall that produces a fall probability.

Stage 3 (Inference Pipeline): ByteTrack associates detections across frames to maintain consistent person identities. Per-track keypoint buffers of up to 30 frames feed the temporal classifier, which produces the final fall probability and activity classification output.

An overview of the complete framework is illustrated in Fig. 3.1, including the shared backbone-neck, the detection and pose heads, ByteTrack-based identity association, temporal keypoint buffering, and the GRU-based temporal action classifier.

3.2 Backbone: MobileViT-S

We employ MobileViT-S [41] as the backbone feature extractor. MobileViT is a hybrid architecture that combines the local feature extraction capabilities of MobileNetV2 [51] inverted residual blocks with the global contextual modeling of Vision Transformers [13]. Specifically, the early stages consist of standard MobileNetV2 blocks that capture local spatial patterns efficiently, while the later stages introduce MobileViT blocks in which feature map patches are processed by lightweight Transformer encoders. This hybrid design enables the network to jointly model local textures and global spatial relationships, which is particularly beneficial for thermal IR imagery where local gradients are subtle and long-range body part dependencies are informative.

MobileViT-S is selected for its favorable accuracy-efficiency trade-off: it achieves competitive performance on ImageNet classification while requiring significantly fewer parameters and FLOPs than full Vision Transformers or heavy CNN backbones, making it well suited

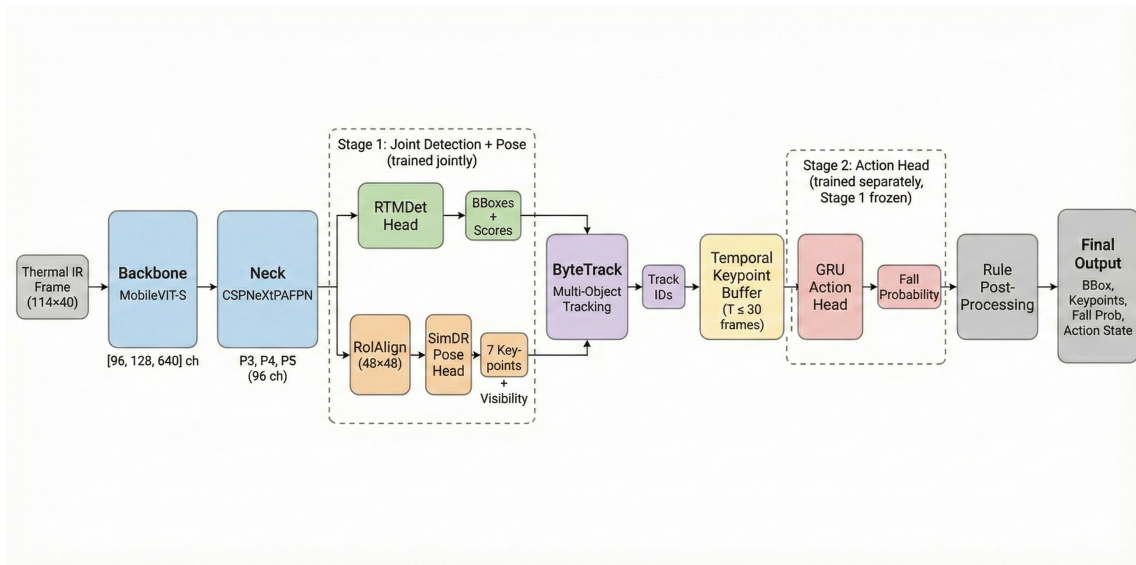


Figure 3.1: Overview of the proposed RTMDetWithPose framework. A shared MobileViT-S backbone and CSPNeXtPAFPN neck produce multi-scale features consumed by both the RTMDet detection head and the SimDR pose estimation head via RoI-aligned features. ByteTrack maintains cross-frame identity association, feeding per-track keypoint sequences to the GRU-based temporal action classifier.

for edge deployment on FPGA-accelerated hardware. We initialize the backbone with ImageNet-pretrained weights from the `timm` library [67] and extract multi-scale features from stages 2, 3, and 4, yielding feature maps with channel dimensions of 96, 128, and 640, respectively.

3.3 Neck: CSPNeXtPAFPN

To fuse multi-scale features from the backbone, we adopt CSPNeXtPAFPN [38], a feature pyramid network that extends the Path Aggregation Feature Pyramid Network (PAFPN) [31] with Cross Stage Partial (CSP) blocks [65] redesigned using the NeXt-style block from RTMDet. The neck receives feature maps with channel dimensions [96, 128, 640] from backbone stages 2, 3, and 4, and produces three output feature maps $\{P_3, P_4, P_5\}$ at strides $\{8, 16, 32\}$, all with a unified channel dimension of 96. The top-down pathway aggregates semantic information from deep layers to shallow layers, while the bottom-up pathway propagates fine-grained spatial details back to deeper levels. We use a single CSP block per fusion stage (`num_csp_blocks=1`) to maintain computational efficiency. The resulting multi-scale feature maps are shared between the detection head and the RoI feature extractor for pose estimation.

3.4 Detection Head: RTMDet

We employ the RTMDet detection head [38], an anchor-free single-stage detector that predicts object locations as distances from feature map grid points to the four bounding box edges. The head operates on all three feature pyramid levels $\{P_3, P_4, P_5\}$ and performs single-class detection (human). Feature channels in the detection head are set to 96, matching the neck output dimension.

Bounding box parameterization. For each spatial location (c_x, c_y) on the feature map, the detector predicts four non-negative distances (l, t, r, b) to the left, top, right, and bottom edges of the bounding box, respectively. The decoded box coordinates are given by:

$$(x_1, y_1, x_2, y_2) = (c_x - l, c_y - t, c_x + r, c_y + b). \quad (3.1)$$

Label assignment. We use the DynamicSoftLabelAssigner with $\text{topk} = 13$, which dynamically assigns positive samples based on a combination of classification confidence and localization quality, replacing the need for fixed IoU-based assignment rules.

Classification loss. The classification branch is trained with Quality Focal Loss (QFL) [27], which jointly models classification confidence and localization quality:

$$\mathcal{L}_{\text{cls}} = -|y - \sigma(\hat{y})|^\beta [y \log \sigma(\hat{y}) + (1 - y) \log(1 - \sigma(\hat{y}))], \quad (3.2)$$

where $\hat{y}$ denotes the predicted logit, $\sigma(\cdot)$ is the sigmoid function, y is the soft target encoding localization quality, and $\beta = 2.0$ controls the focusing strength.

Regression loss. The bounding box regression branch is supervised with Generalized Intersection over Union (GIoU) loss [48]:

$$\mathcal{L}_{\text{bbox}} = 1 - \text{GIoU}(\hat{\mathbf{b}}, \mathbf{b}), \quad (3.3)$$

where $\hat{\mathbf{b}}$ and $\mathbf{b}$ denote the predicted and ground-truth bounding boxes, respectively. The regression loss weight is set to 2.0.

Test-time inference. During inference, detections are filtered with a confidence threshold of 0.3 and refined through non-maximum suppression (NMS) with an IoU threshold of 0.45, retaining at most 100 detections per image.

3.5 Pose Estimation Head: SimDR

We adopt a top-down pose estimation approach in which keypoints are estimated within each detected bounding box. This subsection describes the RoI feature extraction process and the SimDR-based keypoint estimation head.

3.5.1 RoI Feature Extraction

For each detected person, we extract a fixed-size feature representation using RoIAlign [19] applied to the multi-scale feature pyramid $\{P_3, P_4, P_5\}$. The RoIAlign operation produces

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a 48×48 feature map for each person, with bilinear interpolation ensuring sub-pixel alignment. During training, we use ground-truth bounding boxes with $\pm 10\%$ random jitter along each axis to simulate detection noise and improve robustness. During inference, we use the top-K detected bounding boxes ranked by confidence score.

3.5.2 Keypoint Definition

Given the low spatial resolution of thermal IR imagery, we define a reduced skeleton with seven keypoints: (0) Head, (1) Shoulder center, (2) Right hand, (3) Left hand, (4) Hip center, (5) Right foot, and (6) Left foot. This coarse skeleton captures the essential body configuration for fall detection while remaining feasible to estimate from the limited thermal resolution. Each keypoint annotation includes a visibility flag: $v = 0$ (not labeled), $v = 1$ (occluded), and $v \geq 2$ (visible).

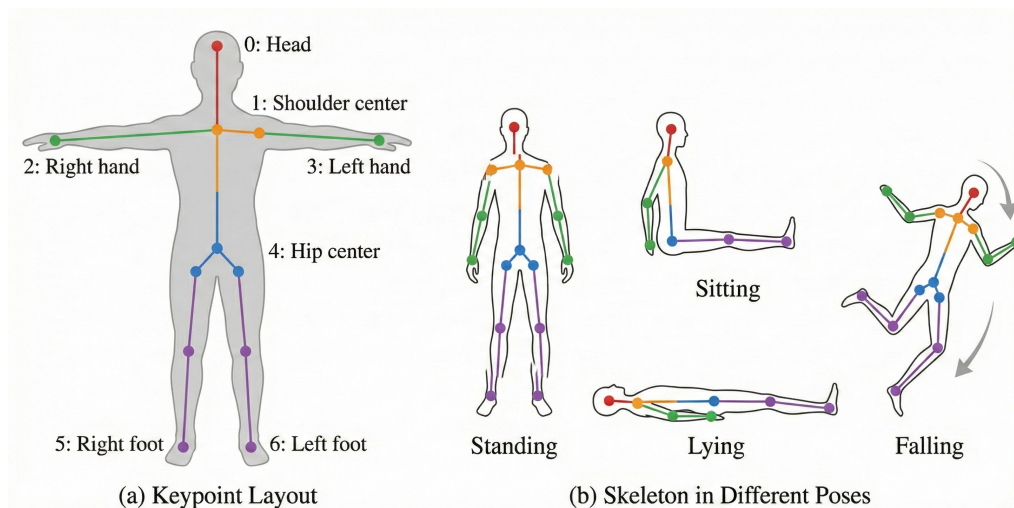


Figure 3.2: Definition of the seven-keypoint skeleton for thermal IR pose estimation. (a) Keypoint layout and bone connections on a representative thermal image. (b) Skeleton configurations across different activity states: standing, sitting, lying, and falling. The reduced skeleton captures essential body structure while remaining estimable at the low resolution of thermopile arrays.

3.5.3 SimDR Architecture

We adopt SimDR (Simple Disentangled Representation) [28], which reformulates two-dimensional keypoint localization as two independent one-dimensional classification problems along the horizontal (x) and vertical (y) axes. This avoids the computational overhead of generating and processing full 2D heatmaps while achieving sub-pixel accuracy through a configurable bin resolution.

Shared feature extractor. The 48×48 RoI-aligned feature map is processed by four convolutional layers, each consisting of a 3×3 convolution (from 96 to 128 channels), batch normalization, and ReLU activation.

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Axis-wise decomposition. The shared feature tensor $\mathbf{F} \in \mathbb{R}^{128 \times H \times W}$ (where $H = W = 48$) is decomposed along each axis via average pooling:

$$\mathbf{f}_x = \frac{1}{H} \sum_{h=1}^H \mathbf{F}[:, h, :] \in \mathbb{R}^{128 \times W}, \quad (3.4)$$

$$\mathbf{f}_y = \frac{1}{W} \sum_{w=1}^W \mathbf{F}[:, :, w] \in \mathbb{R}^{128 \times H}. \quad (3.5)$$

1D classification heads. Each axis-pooled feature is processed by a dedicated one-dimensional head consisting of a 3×1 Conv1d with batch normalization and ReLU, a transposed convolution (ConvTranspose1d with kernel size 4 and stride 2) for $\times 2$ upsampling, and a 1×1 Conv1d that produces logits of dimension $L = W_{\text{feat}} \times k$ for the x -axis and $L = H_{\text{feat}} \times k$ for the y -axis, where $k = 2$ is the SimDR scale factor. With $W_{\text{feat}} = H_{\text{feat}} = 48$ and $k = 2$, each axis produces $L = 96$ classification bins, enabling sub-pixel localization precision.

Visibility branch. A separate branch applies global average pooling over the shared feature map followed by a fully connected layer to predict per-keypoint visibility logits $\hat{v}_j \in \mathbb{R}$ for $j = 1, \dots, 7$.

Coordinate decoding. At inference time, keypoint coordinates are recovered by taking the argmax over the 1D logit vectors and dividing by the scale factor and feature map dimension:

$$\hat{x}_j = \frac{\arg \max(\mathbf{p}_j^x)}{k \cdot W_{\text{feat}}}, \quad \hat{y}_j = \frac{\arg \max(\mathbf{p}_j^y)}{k \cdot H_{\text{feat}}}, \quad (3.6)$$

where $\mathbf{p}_j^x$ and $\mathbf{p}_j^y$ are the softmax-normalized logit vectors for keypoint j along the x - and y -axes, respectively. The resulting coordinates lie in $[0, 1]$ and are subsequently mapped to image coordinates using the bounding box geometry.

To better illustrate the structural difference between conventional heatmap-based pose estimation and the SimDR formulation adopted in this work, we provide a visual comparison in Fig. 3.3. As shown, the traditional 2D Gaussian heatmap approach predicts a dense $K \times H \times W$ tensor for keypoint localization, followed by spatial argmax decoding. In contrast, SimDR decomposes the 2D localization problem into two independent 1D classification tasks along the horizontal and vertical axes via axis-wise pooling. This design substantially reduces output dimensionality and computational overhead while preserving localization precision through fine-grained discretization.

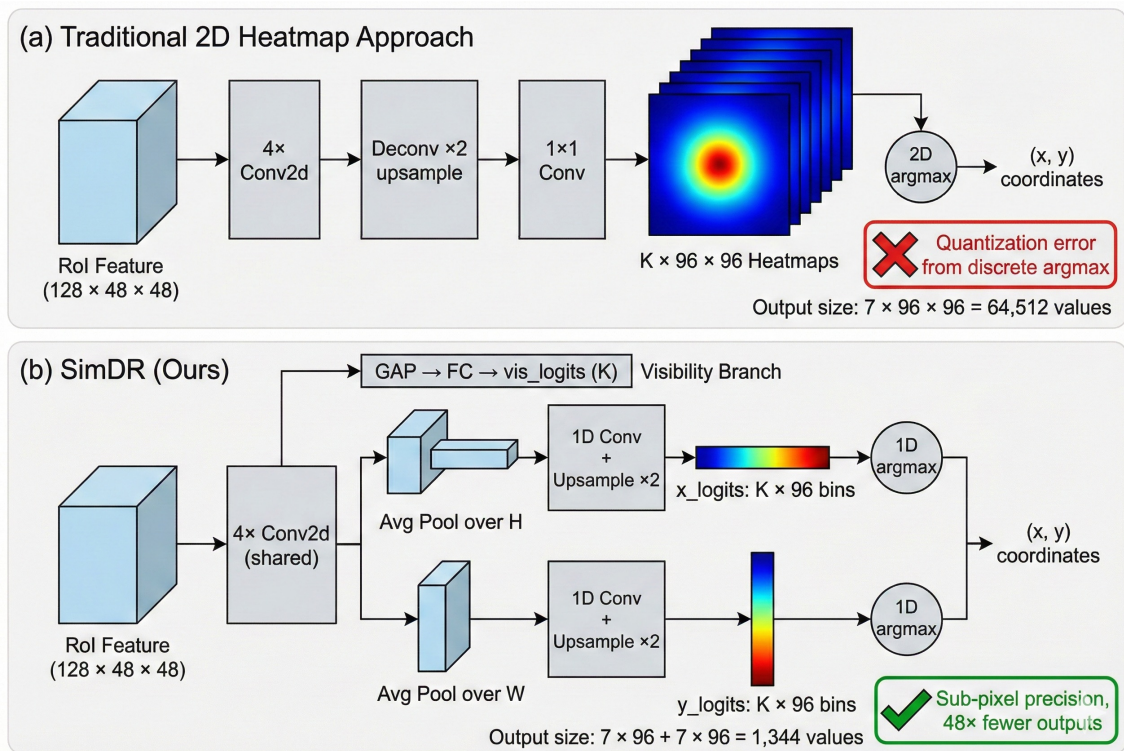


Figure 3.3: Comparison of (a) traditional 2D Gaussian heatmap decoding and (b) the SimDR approach adopted in this work. SimDR decomposes 2D keypoint localization into two independent 1D classification problems along the x - and y -axes via axis-wise average pooling. This reduces the output dimensionality from $K \times H \times W$ to $2 \times K \times L$ while maintaining sub-pixel precision through the scale factor $k = 2$.

3.5.4 Pose Estimation Losses

SimDR coordinate loss. The 1D classification targets are generated as discrete Gaussian distributions centered at the ground-truth bin location with standard deviation $\sigma_{1d} = 1.5$:

$$q_j^x(i) = \frac{\exp\left(-\frac{(i-\hat{i}_j^x)^2}{2\sigma_{1d}^2}\right)}{\sum_{m=1}^L \exp\left(-\frac{(m-\hat{i}_j^x)^2}{2\sigma_{1d}^2}\right)}, \quad (3.7)$$

where $\hat{i}_j^x = \text{round}(x_j^{\text{gt}} \cdot k \cdot W_{\text{feat}})$ is the ground-truth bin index for keypoint j along the x -axis. An analogous target q_j^y is constructed for the y -axis. The SimDR loss is the Kullback–Leibler divergence between the predicted and target distributions, averaged over all visible keypoints:

$$\mathcal{L}_{\text{simdr}} = \frac{1}{K_{\text{vis}}} \sum_{j=1}^K \mathbb{I}[v_j \geq 2] \left[\text{KL}(q_j^x \| p_j^x) + \text{KL}(q_j^y \| p_j^y) \right], \quad (3.8)$$

where $K = 7$ is the number of keypoints, K_{vis} is the count of visible keypoints, and $\mathbb{I}[\cdot]$ is the indicator function.

Visibility loss. The visibility branch is trained with binary cross-entropy loss using three-level weighting: invisible keypoints ($v = 0$) are ignored, occluded keypoints ($v = 1$) receive weight 0.3, and visible keypoints ($v \geq 2$) receive weight 1.0:

$$\mathcal{L}_{\text{vis}} = -\frac{1}{K} \sum_{j=1}^K w_j^v \left[\bar{v}_j \log \sigma(\hat{v}_j) + (1 - \bar{v}_j) \log(1 - \sigma(\hat{v}_j)) \right], \quad (3.9)$$

where $\bar{v}_j$ is the binary visibility target and w_j^v is the visibility-dependent weight.

3.5.5 Alternative Pose Estimation Methods

To validate the choice of SimDR, we implement and compare four additional pose estimation approaches as part of our ablation study:

(A) 2D Gaussian Heatmap: A standard heatmap-based approach [43, 70] that generates a 96×96 heatmap per keypoint via four convolutional layers followed by one deconvolutional layer with $\times 2$ upsampling. The loss is the mean squared error (MSE) between predicted and ground-truth Gaussian heatmaps ($\sigma = 3.0$). Coordinates are extracted via argmax with sub-pixel refinement.

(B) Foreground-Weighted Heatmap with Visibility: Extends method (A) with foreground-weighted MSE [73] that amplifies the loss at keypoint peak locations (foreground weight = 10.0) to mitigate the dominance of background pixels, and an auxiliary visibility branch [25] trained with BCE loss using the three-level weighting scheme described above.

(C) Soft-Argmax / Integral Regression: Extends method (B) with a differentiable soft-argmax (integral) operation [44, 56] that extracts continuous coordinates from the heatmap,

enabling direct supervision of keypoint positions via an additional Smooth L1 loss (weight = 0.5) with temperature parameter $\beta = 10.0$.

(E) Direct Coordinate Regression: A regression baseline [61] that directly predicts (x, y) coordinates from pooled RoI features via fully connected layers, without any intermediate heatmap representation.

3.6 Temporal Action Classification Head

The temporal action classification head is a lightweight GRU-based network that adopts a *dual-task* design: it simultaneously performs four-class activity recognition and binary fall detection from skeleton keypoint sequences. We choose a recurrent architecture over graph convolutional approaches (e.g., ST-GCN [72]) because our reduced seven-keypoint skeleton has a simple spatial structure that does not warrant the additional complexity of graph-based modeling.

3.6.1 Dual-Task Motivation

A naïve approach to fall detection formulates it as a single binary classification problem. However, this fails to exploit the rich activity structure present in the data: falls are inherently *transitions from* a specific activity state (e.g., falling while walking versus falling while standing). We hypothesize that jointly learning to recognize the underlying activity provides the model with a structured representation of normal body dynamics, against which abnormal fall patterns become easier to detect. Concretely, if the model understands what “walking” and “sitting down” look like, it can more reliably identify when a walking sequence degenerates into a fall versus a controlled sitting transition.

Our dual-task formulation therefore comprises two output heads that share a common temporal encoder:

- **Activity head:** four-class classification over coarse activity categories.
- **Fall head:** binary classification for fall / non-fall detection.

3.6.2 Activity Class Definition and Label Mapping

The SensIR dataset provides 10 fine-grained activity labels (1A-1 through 1A-10). We consolidate these into four coarse activity classes that capture the principal body configurations relevant to fall disambiguation:

The four activity classes are defined as: (0) **Standing**—upright postures including standing still, standing up, and getting up; (1) **Walking**—ambulatory motion; (2) **Transition**—controlled postural changes such as sitting down; and (3) **Lying**—recumbent postures including lying down. Notably, fall samples retain their *pre-fall activity class* (e.g., “falling while walking” is labeled as Walking + Fall), enabling the model to learn activity-conditioned fall patterns. The binary fall label is derived independently: classes 1A-7 through 1A-10 are mapped to $y_{\text{fall}} = 1$, and all others to $y_{\text{fall}} = 0$.

Table 3.1: Mapping from the 10 original SensIR activity labels to four coarse activity classes and the binary fall label used for dual-task training.

Original Label	Activity (4-cls)	Fall
1A-1 Standing still	0: Standing	0
1A-4 Standing up	0: Standing	0
1A-6 Getting up	0: Standing	0
1A-2 Walking	1: Walking	0
1A-3 Sitting down	2: Transition	0
1A-5 Lying down	3: Lying	0
1A-7 Falling while walking	1: Walking	1
1A-8 Falling while standing	0: Standing	1
1A-9 Falling while sitting	2: Transition	1
1A-10 Falling and standing up	0: Standing	1

3.6.3 Input Feature Representation

For each tracked person at frame t , we construct a 35-dimensional feature vector by concatenating five features per keypoint:

$$\mathbf{f}_t = [x_1, y_1, v_1, \Delta x_1, \Delta y_1, \dots, x_7, y_7, v_7, \Delta x_7, \Delta y_7] \in \mathbb{R}^{35}, \quad (3.10)$$

where (x_j, y_j) are keypoint coordinates normalized to $[0, 1]$ relative to the bounding box, v_j is the visibility score from the SimDR head, and $(\Delta x_j, \Delta y_j) = (x_j^{(t)} - x_j^{(t-1)}, y_j^{(t)} - y_j^{(t-1)})$ are frame-to-frame velocity features. Keypoints that lie outside the bounding box have their coordinates and visibility set to zero. The temporal input is a sequence $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_T] \in \mathbb{R}^{T \times 35}$, where $T = 8$ during training and $T \leq 30$ during inference.

Sequence padding and cold-start. During training, when a person’s track has fewer than $T = 8$ frames of history, the sequence is padded to length 8 by repeating the first available frame. During inference, the GRU processes variable-length sequences from $T = 1$ without padding; for the first frame, velocity features are initialized to zero ($\Delta x_j = \Delta y_j = 0$), and the GRU updates its hidden state incrementally as new frames arrive.

3.6.4 Per-Keypoint Importance Weighting

Prior to encoding, we apply learned importance weights to each keypoint’s feature block. The weight vector is defined as $\mathbf{w} = [1.5, 1.0, 0.5, 0.5, 1.5, 1.2, 1.2]$ for Head, Shoulder, Right hand, Left hand, Hip, Right foot, and Left foot, respectively. The Head and Hip keypoints receive the highest weights (1.5) as they are most discriminative for fall direction; the Feet receive moderate weights (1.2) for motion detection; and the Hands receive low weights (0.5) as they are least informative for fall classification.

3.6.5 Dual-Task Network Architecture

The temporal action classifier comprises a shared encoder–GRU trunk and two task-specific output heads:

Per-frame encoder. A two-layer MLP transforms the 35-dimensional input into a 64-dimensional per-frame representation:

$$\mathbf{h}_t = \text{ReLU}(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \mathbf{f}'_t + \mathbf{b}_1) + \mathbf{b}_2), \quad (3.11)$$

where $\mathbf{W}_1 \in \mathbb{R}^{64 \times 35}$, $\mathbf{W}_2 \in \mathbb{R}^{64 \times 64}$, and dropout ($p = 0.2$) is applied after the first ReLU.

Temporal GRU. A single-layer GRU [9] with hidden size 64 processes the sequence of per-frame features, followed by Layer Normalization [1]:

$$\mathbf{g}_T = \text{LayerNorm}(\text{GRU}(\mathbf{h}_1, \dots, \mathbf{h}_T)), \quad (3.12)$$

where $\mathbf{g}_T \in \mathbb{R}^{64}$ is the final hidden state.

Temporal residual connection. To stabilize early training when GRU weights are randomly initialized, we add a skip connection from the last-frame encoder output:

$$\mathbf{o} = \mathbf{h}_T + \mathbf{g}_T. \quad (3.13)$$

Activity head. A linear layer maps the shared feature to four-class activity logits:

$$\mathbf{z}_{\text{act}} = \mathbf{W}_{\text{act}} \mathbf{o} + \mathbf{b}_{\text{act}} \in \mathbb{R}^4, \quad (3.14)$$

with predicted class $\hat{c} = \arg \max(\mathbf{z}_{\text{act}})$.

Fall head. A separate linear layer maps the same shared feature to a scalar fall logit:

$$p_{\text{fall}} = \sigma(\mathbf{w}_{\text{fall}}^T \mathbf{o} + b_{\text{fall}}), \quad (3.15)$$

where $\sigma(\cdot)$ denotes the sigmoid function. During inference, a fall alarm is triggered when $p_{\text{fall}} \geq \tau$, with decision threshold $\tau = 0.5$.

Design rationale. The two heads share the encoder and GRU but maintain separate linear projections. This parameter-efficient design ensures that the shared temporal representation captures both activity-discriminative features (spatial pose patterns distinguishing standing from walking from lying) and fall-discriminative features (temporal dynamics of rapid descent and deceleration). The activity head provides an auxiliary learning signal that regularizes the shared representation, preventing it from collapsing to fall-only features. At inference time, the activity prediction provides an interpretable context for each fall alarm (e.g., “fall detected during walking”), aiding clinical decision-making.

3.6.6 Dual-Task Loss Function

The dual-task action head is trained with a combined loss comprising an activity classification loss and a focal binary cross-entropy fall detection loss.

Activity classification loss. The four-class activity head is supervised with label-smoothed cross-entropy loss:

$$\mathcal{L}_{\text{act}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=0}^3 \tilde{y}_{i,c} \log \frac{\exp(z_{i,c})}{\sum_{c'} \exp(z_{i,c'})}, \quad (3.16)$$

Fig 5: GRU-based Temporal Action Classification Head

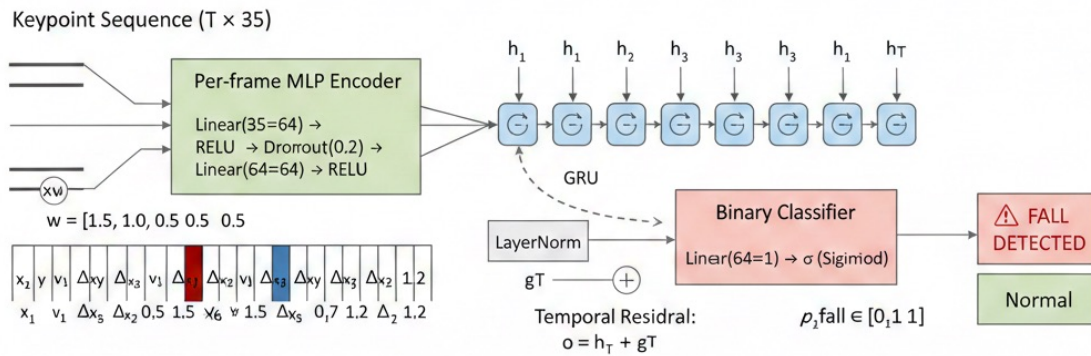


Figure 3.4: Architecture of the dual-task GRU-based temporal action classification head. A shared per-frame encoder and GRU temporal model produce a fused feature o , which is consumed by two task-specific heads: a four-class activity classifier and a binary fall detector. The shared trunk learns a joint representation that captures both normal activity patterns and abnormal fall dynamics.

where $\tilde{y}_{i,c} = (1 - \epsilon) y_{i,c} + \epsilon/4$ denotes the label-smoothed target with smoothing factor $\epsilon = 0.1$. Label smoothing prevents overconfident activity predictions, which is beneficial given the inherent ambiguity between adjacent activity states (e.g., the transition boundary between standing and sitting).

Focal fall detection loss. The binary fall head is trained with Focal Binary Cross-Entropy loss [29] to address the severe class imbalance between fall and non-fall samples:

$$\mathcal{L}_{\text{focal}}(p, y) = -(1 - p_t)^\gamma \log(p_t), \quad (3.17)$$

where $p_t = p$ if $y = 1$ (fall) and $p_t = 1 - p$ otherwise, with focusing parameter $\gamma = 2.0$. The focal modulating factor $(1 - p_t)^\gamma$ down-weights easy, well-classified examples and concentrates learning on hard, ambiguous cases near the decision boundary.

Per-sample importance weighting for falls. We apply sample-dependent weights based on the activity class to address inter-class confusion:

$$w_i = \begin{cases} w_{\text{pos}} = 2.0 & \text{if } y_{\text{fall},i} = 1 \text{ (falling),} \\ w_{\text{hard}} = 3.0 & \text{if } y_{\text{fall},i} = 0 \text{ and } \hat{c}_i \in \{2, 3\}, \\ 1.0 & \text{otherwise,} \end{cases} \quad (3.18)$$

where $\hat{c}_i$ is the activity class. Transition ($c = 2$) and Lying ($c = 3$) samples are designated as *hard negatives* because they involve body configurations highly similar to falling but are not actual falls. Assigning them a weight of 3.0 forces the model to learn the subtle kinematic distinctions—primarily the speed of descent and limb dynamics—that differentiate controlled postural transitions from uncontrolled falls.

Combined dual-task loss. The final action loss balances the two objectives:

$$\mathcal{L}_{\text{action}} = \lambda_{\text{fall}} \cdot \frac{1}{N} \sum_{i=1}^N w_i (1 - p_{t,i})^\gamma \text{BCE}(\hat{y}_i, y_{\text{fall},i}) + \lambda_{\text{act}} \cdot \mathcal{L}_{\text{act}}, \quad (3.19)$$

with $\lambda_{\text{fall}} = 2.0$ and $\lambda_{\text{act}} = 1.0$. The fall loss receives a higher weight to ensure that fall detection—the primary clinical objective—remains the dominant learning signal, while the activity loss serves as a complementary regularizer that enriches the shared temporal representation.

3.7 Multi-Object Tracking: ByteTrack

During video inference, we employ ByteTrack [74] to associate detections across frames and maintain consistent person identities. ByteTrack uses a cascaded two-stage association strategy.

Motion model. A Kalman filter [2] predicts each tracklet's bounding box in the next frame based on a constant-velocity motion model.

First association. High-confidence detections (score > 0.6) are matched to confirmed tracks using IoU-based cost (threshold = 0.1) via the Hungarian algorithm.

Second association. Unmatched high-confidence detections are matched to tentative tracks (IoU threshold = 0.3).

Third association. Low-confidence detections (score in $[0.1, 0.6]$) are matched to remaining unmatched tracks (IoU threshold = 0.5), recovering partially occluded persons that would otherwise be discarded.

Track lifecycle. New tracks are initialized from unmatched detections with confidence > 0.7 and remain tentative for 3 frames before confirmation. Tracks without matches for 30 consecutive frames are deleted. Each active track maintains a keypoint buffer of up to $T = 30$ frames, from which the temporal action classifier draws its input sequence.

3.8 Multi-Task Training Strategy

3.8.1 Two-Stage Training

The model is trained in two sequential stages to ensure stable multi-task optimization:

Stage 1: Joint detection and pose estimation. The backbone (MobileViT-S), neck (CSP-NeXtPAFPN), detection head (RTMDet), and pose head (SimDR) are trained jointly end-to-end. The combined loss is:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{cls}} + 2.0 \cdot \mathcal{L}_{\text{bbox}} + 5.0 \cdot \mathcal{L}_{\text{simdr}} + 1.0 \cdot \mathcal{L}_{\text{vis}}. \quad (3.20)$$

Training uses the AdamW optimizer [36] with learning rate 2×10^{-4} , weight decay 10^{-4} , and gradient clipping at 10. The learning rate is warmed up linearly over 1000 iterations (start factor = 0.001) and then decayed according to a cosine annealing schedule [37]. Training runs for approximately 75 epochs with batch size 4.

Stage 2: Action head training with frozen Stage 1. All parameters trained in Stage 1 are frozen, and only the randomly initialized ActionTemporalHead is optimized. The loss is $\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{action}}$. Training uses AdamW with learning rate 10^{-4} , weight decay 10^{-3} , linear warmup over 500 iterations, and cosine annealing. The best checkpoint is selected based on fall F1-score on a validation set.

The two-stage strategy is motivated by two considerations. First, the detection and pose estimation tasks require substantially more epochs to converge than the action head, which is a comparatively small network. Second, freezing the detection and pose parameters prevents catastrophic forgetting of well-learned visual features when the action head's gradients flow backward through the shared representation during joint optimization. During Stage 2, the action head receives clean keypoint inputs from the pre-trained pose estimator, ensuring stable and efficient training.

3.8.2 Data Augmentation

The training pipeline applies the following augmentations: random rotation ($\pm 20^\circ$, $p = 0.5$) with synchronized keypoint coordinate transformation; resize to 384×384 (without preserving aspect ratio); random horizontal flip ($p = 0.5$); and photometric distortion

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(brightness ± 32 , contrast in $[0.5, 1.5]$, saturation in $[0.5, 1.5]$, hue ± 18). For the action head, temporal keypoint sequences of length $T = 8$ are provided as pre-computed fields in the dataset, with each frame's feature vector containing normalized coordinates, visibility scores, and velocity displacements.

Chapter 4 Experiment

This chapter presents a comprehensive experimental evaluation of the proposed thermal infrared fall detection system. We first describe the dataset and evaluation protocol (Sections 4.1–4.2), followed by implementation details (Section 4.3). We then conduct a series of ablation studies to validate our design choices: pose estimation representation (Section 4.4), fall classification strategy (Section 4.5), loss function design (Section 4.6), and input feature representation (Section 4.7). Finally, we present the full system comparison (Section 4.8), qualitative results (Section 4.9), computational cost analysis (Section 4.10), and a consolidated discussion (Section 4.11).

4.1 Dataset

The thermal infrared fall detection dataset used in this study was collected using a Heimann HTPA 60×40 thermopile sensor array within the SensIR project. The dual-sensor configuration yields a combined input resolution of approximately 114 × 40 pixels, resized to 384 × 384 for training and inference. Annotations follow the COCO format and include: (i) bounding boxes for a single “human” class in $[x, y, w, h]$ format; (ii) seven keypoints per person—Head (0), Shoulder center (1), Right hand (2), Left hand (3), Hips center (4), Right foot (5), and Left foot (6)—each with a three-level visibility flag (0 = out-of-frame/unlabeled, 1 = occluded, 2 = visible); (iii) a ten-class action label per instance; and (iv) pre-computed temporal keypoint sequences of $T = 8$ frames with normalized keypoint features (five dimensions per keypoint: x, y , visibility, $\Delta x, \Delta y$).

The ten action classes comprise six non-falling categories (Standing still, Walking, Sitting down, Standing up, Lying down, Getting up) and four falling categories (Falling while walking, Falling while standing, Falling while sitting, Falling while standing up). For binary fall detection evaluation, classes 6–9 are grouped as “falling” (positive) and classes 0–5 as “not falling” (negative). Table 4.1 summarizes the dataset statistics. The training split contains 5,104 images with 4,838 annotated instances, of which 1,482 (30.6%) are falling events. This relatively modest dataset size—combined with a ~70:30 non-falling to falling class imbalance—imposes important constraints on model capacity, generally favoring simpler architectures over high-capacity designs. Moreover, several non-falling actions, particularly “Lying down” and “Getting up,” share highly similar body configurations with actual falling events. This visual ambiguity motivates our hard negative mining strategy, which assigns elevated loss weights to these confusable classes during training. Finally, compared to RGB imagery, thermal infrared images exhibit substantially lower spatial resolution and contrast, which renders fine-grained keypoint localization more challenging.

All frame-level metrics are reported on the validation split. Additionally, a separate sequence-level test set comprising 38 video sequences (18 fall, 20 non-fall) from six activity scenarios is used to evaluate end-to-end detection performance at the event level.

Table 4.1: Dataset statistics.

Split	Images	Instances	Falling (cls 6–9)	Non-falling (cls 0–5)	Fall Ratio
Train	5,104	4,838	1,482	3,356	30.6%
Val	866	861	229	632	26.6%

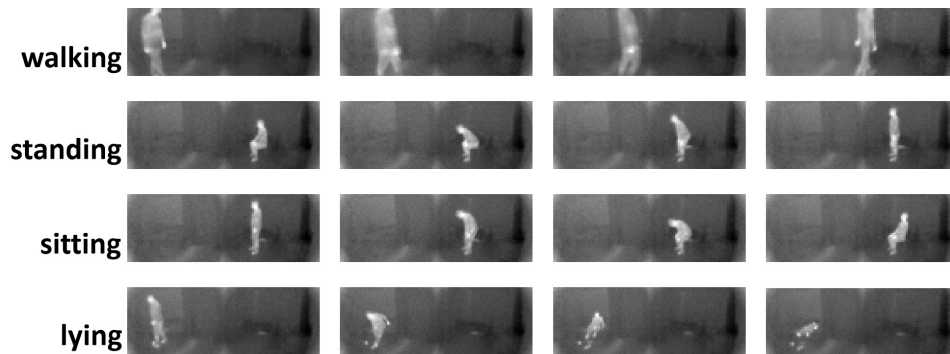


Figure 4.1: Qualitative examples of thermal IR sequences. Columns indicate pose categories

4.2 Evaluation Metrics

We evaluate the system at three levels. **Detection** performance is measured using the standard COCO bounding-box mAP@[.50:.95] and COCO Keypoint AP based on Object Keypoint Similarity (OKS). Since our keypoint set is non-standard (seven keypoints rather than the COCO 17), OKS sigmas are uniformly set to 0.25 for all keypoints. **Posture classification**, applicable only to the proposed dual-task method, is reported as four-class accuracy and per-class F1 for the posture categories {standing, walking, sitting, lying}. Instance-level matching employs an IoU threshold of 0.5 between predicted and ground-truth boxes. **Fall detection** is evaluated as a binary classification task: classes 6–9 constitute the positive (falling) class, while classes 0–5 form the negative class. We report binary precision, recall, F1-score, and Average Precision (AP). The primary metric for model selection is **Fall F1**, since the clinical application prioritizes correctly identifying falls. Fall AP provides a complementary threshold-independent assessment.

4.3 Implementation Details

All experiments were conducted on a single NVIDIA TITAN RTX GPU using the MMDetection 3.x framework [6]. Training proceeds in two stages.

Stage 1: Joint Detection and Pose Estimation. The backbone is MobileViT-S [41], initialized with ImageNet-pretrained weights from the timm library [67]. The neck is a CSPNeXtPAFPN with input channels [96, 128, 640] and output channels 96. Input images are resized to 114×40 pixels. We use the AdamW optimizer [36] with a learning rate of 2×10^{-4} , weight decay of 1×10^{-4} , and gradient clipping at norm 10. The learning rate schedule comprises a linear warmup over the first 1,000 iterations followed by cosine annealing [37]. Training runs for 75 epochs with a batch size of 4. The multi-task loss

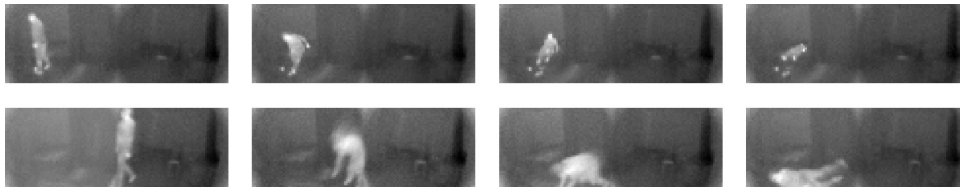


Figure 4.2: Qualitative examples illustrating the visual ambiguity between lying and falling in thermal infrared sequences.

weights are: Quality Focal Loss [27] for classification ($\lambda_{\text{cls}} = 1.0$), GIoU loss [48] for box regression ($\lambda_{\text{bbox}} = 2.0$), KL divergence for SimDR [28] keypoint localization ($\lambda_{\text{kl}} = 5.0$), and BCE for visibility prediction ($\lambda_{\text{vis}} = 1.0$). RoI features are extracted via RoIAlign [19] at an output size of 48×48 from multi-scale feature maps (strides $\{8, 16, 32\}$), with $\pm 10\%$ random jitter applied to ground-truth boxes during training. Data augmentation includes random rotation ($\pm 20^\circ$, $p = 0.5$), horizontal flipping ($p = 0.5$), and photometric distortion (brightness ± 32 , contrast 0.5–1.5, saturation 0.5–1.5, hue ± 18).

Stage 2: Temporal Action Head. All Stage 1 parameters (backbone, neck, detection head, pose head) are frozen, and the best Stage 1 checkpoint is loaded (bbox mAP = 0.866, Keypoint AP = 0.802). Only the randomly initialized temporal action head is trainable. We use AdamW with a reduced learning rate of 1×10^{-4} , weight decay of 1×10^{-3} , and the same warmup-plus-cosine schedule (500-iteration warmup). Batch size is 4 and gradient clipping is applied at max norm 10. Model selection is based on the best validation Fall F1.

Inference Pipeline. ByteTrack [74] is employed for multi-object tracking with high-score threshold 0.6, low-score threshold 0.1, initialization threshold 0.7, and a retention window of 30 frames. Non-maximum suppression uses a score threshold of 0.3 and an IoU threshold of 0.45. A temporal buffer of up to 30 frames per tracked person feeds keypoint sequences into the GRU-based action classifier.

4.4 Ablation Study: Pose Estimation Representation

This ablation compares three pose estimation approaches for the Stage 1 pose head. All variants share the same backbone (MobileViT-S), neck (CSPNeXtPAFPN), and RoI extraction (48×48). The key difference lies in how the extracted RoI feature is decoded into keypoint coordinates.

4.4.1 2D Heatmap

The 2D heatmap baseline generates K two-dimensional heatmaps of size $H' \times W'$, upsampled by a factor of 2 via transposed convolution. Each keypoint location is decoded via argmax over the spatial dimensions. Training uses a foreground-weighted MSE loss:

$$\mathcal{L}_{\text{heatmap}} = \frac{1}{K} \sum_{k=1}^K w_k \cdot \text{MSE}_{\text{fg}}(\hat{H}_k, H_k^{\text{gt}}), \quad (4.1)$$

Table 4.2: Comparison of pose estimation methods (Stage 1, 75–80 training epochs).

Method	Loss	Kpt AP	bbox mAP	bbox AP@0.5
2D Heatmap	FG-MSE + vis BCE	0.573	0.897	0.994
Direct Regression	SmoothL1 + vis BCE	0.570	0.875	0.979
SimDR (Adopted)	KL div + vis BCE	0.802	0.866	0.989

where MSE_{fg} applies a $10\times$ weight to foreground pixels (Gaussian peak region, $H_k^{gt} > 0.01$) relative to background, and the Gaussian target has $\sigma = 2.0$. Keypoint visibility is predicted by a separate branch (global average pooling followed by a fully connected layer) with BCE loss. The architecture consists of four convolutional layers ($96 \rightarrow 128$ channels, each with batch normalization and ReLU) followed by a transposed convolution for $2\times$ upsampling and a 1×1 convolution producing K heatmap channels.

4.4.2 Direct Coordinate Regression

Instead of generating 2D spatial maps, the direct regression approach produces normalized coordinates $(\hat{x}, \hat{y}) \in [0, 1]$ within the RoI through fully connected layers. The loss is:

$$\mathcal{L}_{\text{reg}} = \sum_{k=1}^K w_k \cdot \text{SmoothL1}(\hat{\mathbf{p}}_k, \mathbf{p}_k^{\text{gt}}), \quad (4.2)$$

where $\hat{\mathbf{p}}_k$ and $\mathbf{p}_k^{\text{gt}}$ denote predicted and ground-truth keypoint coordinates, respectively. The architecture employs four convolutional layers ($96 \rightarrow 128$, with batch normalization and ReLU), followed by global average pooling and two fully connected layers ($128 \rightarrow 256 \rightarrow 256$). Two parallel output heads produce coordinates (via sigmoid activation, $K \times 2$ outputs) and visibility logits (K outputs). Coordinate outputs are initialized to $\text{sigmoid}(0) = 0.5$, corresponding to the RoI center, to facilitate faster convergence.

4.4.3 SimDR (Adopted)

SimDR [28] reformulates 2D keypoint localization as two independent 1D classification tasks along the x - and y -axes, using KL divergence against 1D Gaussian targets ($\sigma = 1.5$) and a scale factor $s = 2$ yielding 96 bins per axis for sub-pixel precision. The full architecture and loss formulation are described in Section 3.5.

4.4.4 Pose Estimation Results

Table 4.2 presents the quantitative comparison across all three pose estimation methods trained for 75–80 epochs.

SimDR dramatically outperformed both alternative approaches on the primary Keypoint AP metric, achieving 0.802 compared to 0.573 for the 2D Heatmap and 0.570 for Direct Regression—a relative improvement of approximately 40%. The advantage became even more pronounced at stricter OKS thresholds: SimDR attained an AP@0.75 of 0.874, compared to 0.574 and 0.595 for the 2D Heatmap and Direct Regression methods,

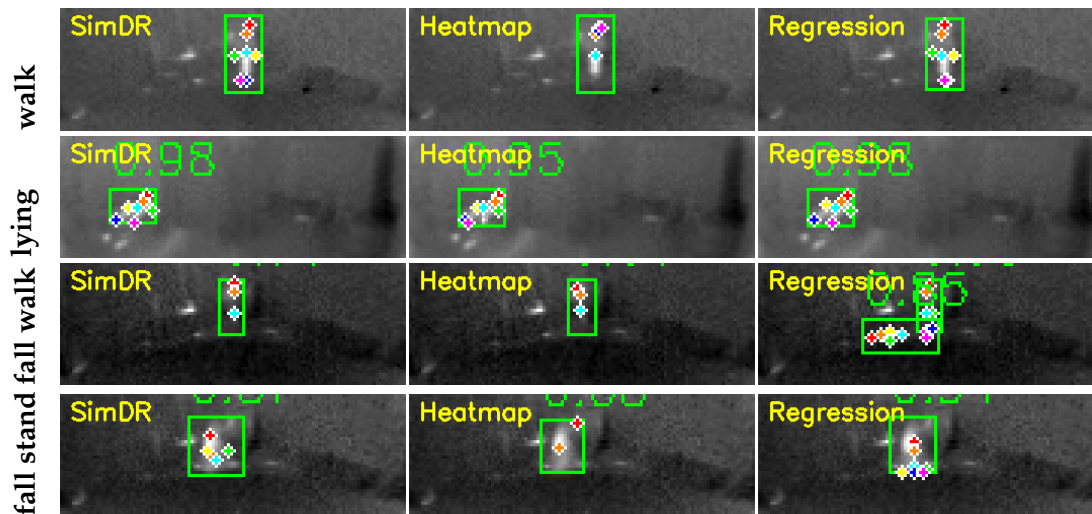


Figure 4.3: Qualitative comparison of pose heads on thermal IR examples. Each row shows one example; each image contains three columns: SimDR, Heatmap, and Regression.

respectively. This substantial gap demonstrates that SimDR produces more precisely localized keypoints, not merely more frequently detected ones.

Detection performance remained comparable across all three variants (bbox mAP ranging from 0.866 to 0.897), confirming that the pose head choice does not adversely affect the shared detection head. The 2D Heatmap variant achieved slightly higher bbox mAP (0.897 vs. 0.866), which we attribute to the heatmap-based loss providing additional spatial regularization to the shared backbone features. However, this marginal detection advantage is far outweighed by SimDR’s substantial pose estimation superiority.

The success of SimDR can be attributed to two factors. First, the 1D classification formulation avoids the quantization error inherent in 2D heatmap decoding, where performing argmax over a broad Gaussian peak on a low-resolution 48×48 feature map introduces localization ambiguity. SimDR’s disentangled 1D distributions, with 96 bins per axis, provide finer spatial granularity. Second, the KL divergence loss against soft 1D Gaussian targets yields smoother gradient landscapes than either hard-argmax MSE (used by the 2D Heatmap) or SmoothL1 regression. Direct Regression, meanwhile, suffers from the well-known difficulty of mapping the entire $[0, 1]$ coordinate range through a sigmoid activation, which produces vanishing gradients near extreme values and provides less forgiving error signals for small localization offsets.

While quantitative metrics provide a global comparison, they do not fully reflect spatial stability under challenging low-resolution thermal conditions. Therefore, Fig. 4.3 presents representative qualitative examples across four typical scenarios: walking, lying, fall while walking, and fall from standing.

As shown in Fig. 4.3, the SimDR head generally produces more spatially consistent and anatomically plausible keypoint layouts, particularly in fall-related frames where body contours are less distinct. In contrast, the heatmap and regression variants exhibit increased

Table 4.3: Comparison of fall classification strategies (Stage 2, frozen SimDR backbone).

Method	Architecture	Fall F1	Fall Prec	Fall Recall	Fall AP
Proposed (Dual-task)	4-class + binary	0.669	1.000	0.502	0.725
Single-task Binary	Binary only	0.667	0.808	0.568	0.744

localization drift in low-contrast regions. This observation is consistent with the significant AP improvement reported in Table 4.2.

4.5 Ablation Study: Fall Classification Strategy

This ablation compares two action head designs for Stage 2, both using the same frozen Stage 1 backbone (SimDR pose head, bbox mAP = 0.866, Keypoint AP = 0.802).

4.5.1 Proposed: Dual-Task Posture Classification and Fall Detection

The proposed design is described in full in Section 3.6. In brief, it employs a shared GRU temporal encoder with two output heads: a **four-class posture head** (standing, walking, sitting, lying) trained with label-smoothed cross-entropy, and a **binary fall head** trained with focal BCE and hard negative weighting (3× for confusable Transition/Lying classes). The fall label is derived independently from the activity label: annotation classes 6–9 map to $y_{\text{fall}} = 1$, classes 0–5 to $y_{\text{fall}} = 0$.

4.5.2 Single-Task Binary Fall Detection (Ablation)

This variant focuses all model capacity on binary fall detection, using the same GRU encoder but with a single binary classifier head and no posture branch. It retains the same focal loss, hard negative mining, and per-keypoint weighting as the proposed method. The loss is:

$$\mathcal{L}_{\text{binary}} = \lambda \cdot \frac{1}{N} \sum_{i=1}^N w_i \cdot (1 - p_{t,i})^\gamma \cdot \text{BCE}(\hat{y}_i, y_i). \quad (4.3)$$

All other hyperparameters are identical to those of the proposed method.

4.5.3 Fall Classification Results

Table 4.3 compares the two classification strategies, and Table 4.4 provides the per-class posture metrics of the proposed method.

The proposed dual-task method achieved comparable Fall F1 (0.669) to the single-task binary variant (0.667), while additionally providing interpretable posture classification output. Most notably, the proposed method achieved *perfect fall precision* (1.000), indicating that every predicted fall was a true fall—there were zero false alarms. The trade-off manifested in recall (0.502 vs. 0.568), suggesting the model adopted a conservative decision boundary. From a clinical deployment perspective, the absence of false positives is highly desirable: when the system triggers a fall alert, care staff can fully trust it.

Table 4.4: Proposed method: per-class posture classification metrics.

Class	Precision	Recall	F1	Support
Standing	0.387	0.189	0.254	370
Walking	0.190	0.473	0.272	110
Sitting	0.056	0.288	0.094	52
Lying	0.864	0.368	0.516	329
Macro-F1	—	—	0.284	861

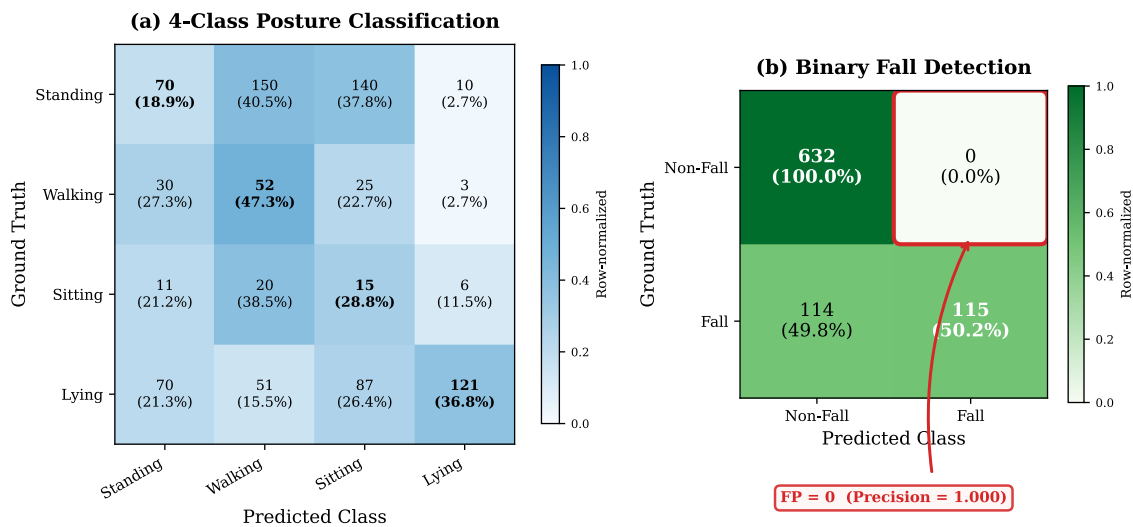


Figure 4.4: Confusion matrices of the proposed dual-task method. (a) 4-class posture classification confusion matrix (row-normalized). (b) Binary fall detection confusion matrix. The proposed method achieves **zero false positives** (Precision = 1.000), ensuring that every fall alert is a true event.

Examining the posture head in detail, the “lying” class achieved the highest F1 (0.516) with strong precision (0.864), meaning the system could reliably detect when a person was in a lying position—a capability critical for post-fall monitoring. The remaining posture classes exhibited lower performance (macro-F1 = 0.284), limited by the small training set and the inherent confusion between standing/walking transitions and between sitting/lying in low-resolution thermal imagery. Nevertheless, the posture output represents clinically valuable information that single-task approaches entirely lack.

Both methods were fundamentally constrained by the dataset size. With only 4,838 training instances, the dual-task model’s higher capacity could not be fully utilized. We anticipate that with synthetic data augmentation (Section 4.8), the proposed method’s richer representation will yield a clearer advantage.

Table 4.5: Loss function comparison (binary fall detection, single-task).

Method	Focal	Hard Neg	Kpt Wt	Fall F1	Fall Prec	Fall AP
Binary-BCE (baseline)	No	No	No	0.748	0.923	0.780
Binary-BCE + anti-overfit	No	No	No	0.520	0.529	0.657
Binary-BCE + mild reg.	No	No	No	0.665	1.000	0.666
Binary-Focal (full)	Yes ($\gamma=2$)	Yes ($3\times$)	Yes	0.667	0.808	0.744

4.6 Ablation Study: Loss Function and Training Strategy

This section evaluates the effect of different loss functions on the single-task binary GRU architecture with raw keypoint features, isolating the contribution of each component.

Table 4.5 reveals that the straightforward Binary-BCE baseline achieved the highest Fall F1 (0.748) on the current dataset, with strong precision (0.923) and the best recall (0.629). This suggests that on a small dataset, simple BCE with a positive weight ($\text{pos_weight} = 3.0$) provides effective class-imbalance handling without overcomplicating the optimization landscape.

Introducing focal loss and hard negative mining slightly reduced overall F1 (0.667 vs. 0.748), primarily by trading recall for a more conservative decision threshold. The focal loss, by down-weighting well-classified examples, caused the model to focus disproportionately on ambiguous samples near the decision boundary. While this mechanism is designed to improve performance on hard examples, it became counterproductive on our limited training set where there were insufficient hard examples for the model to learn robust discriminative patterns.

Over-regularization proved clearly detrimental. The anti-overfit variant—with increased weight decay and dropout—dramatically degraded performance (F1: 0.520), indicating that the baseline model was not actually overfitting with standard regularization. The mild regularization variant achieved perfect precision (1.000) but very low recall (0.498), symptomatic of an overly conservative decision boundary that classified most borderline cases as non-falls. These results confirm that the advanced loss components (focal loss, hard negative mining, per-keypoint weighting) represent architectural investments that will pay off with more training data, rather than universally beneficial modifications.

4.7 Ablation Study: Input Feature Representation

This section compares different input feature representations for the temporal action head. All variants use the same GRU architecture and frozen SimDR backbone.

Table 4.6 demonstrates that raw keypoint features decisively outperformed bone skeleton features (Fall F1: 0.748 vs. 0.680). This result reflects the importance of *absolute position information* for fall detection. A person's head being located in the lower portion of their bounding box ($y > 0.7$) constitutes a strong fall signal, and this spatial cue is lost when using translation-invariant bone vectors that encode only relative joint displacements. While bone representations offer theoretical advantages for pose-invariant action recognition

Table 4.6: Feature representation comparison (all using GRU architecture, frozen SimDR backbone).

Method	Feature Type	Input Dim	Fall F1	Fall Prec	Fall Recall	Fall AP
Skeleton-GRU	Bone vectors	36	0.680	0.713	0.651	0.743
Skeleton+Appearance	Bone + RoI GAP	132	0.641	0.577	0.721	0.683
Binary-BCE (keypoint)	Raw keypoints	35	0.748	0.923	0.629	0.780

Table 4.7: Complete system comparison. bbox mAP and Kpt AP are evaluated on the frozen Stage 1 checkpoint integrated into the full pipeline and therefore differ from the standalone Stage 1 results in Table 4.2.

Method	Pose	Classifier	Feature	Fall F1	Fall Prec	Fall AP	bbox mAP	Kpt AP
Skel.-GRU	SimDR	10-cls CE	Bone (36d)	0.680	0.713	0.743	0.933	0.736
Skel.+App.	SimDR	10-cls CE	Bone+RoI	0.641	0.577	0.683	0.934	0.746
Bin.-BCE	SimDR	Bin. BCE	Kpt (35d)	0.748	0.923	0.780	0.933	0.741
Proposed	SimDR	4-cls+bin.	Kpt (35d)	0.669	1.000	0.725	0.931	0.734
Prop.+Synth.	SimDR	4-cls+bin.	Kpt (35d)	0.776	0.951	0.834	0.931	0.734

in RGB settings, the fall detection task fundamentally depends on the body's spatial configuration relative to the ground plane.

Adding appearance features to the bone representation actually degraded performance (Skeleton+Appearance F1: 0.641 vs. Skeleton-GRU F1: 0.680). The 96-dimensional RoI global average pooling features introduced noise because thermal infrared appearance is inherently low-contrast and highly variable—providing little discriminative information beyond what the skeleton already captures. Furthermore, the 3.7× increase in input dimensionality (from 36 to 132) exacerbated overfitting on the limited training set.

The Multi-task Auxiliary approach, which employed three competing heads (10-class action, 3-class state, and binary fall) sharing a single GRU encoder with bone features, suffered severe underfitting (F1: 0.249). Despite achieving high precision (0.917), its recall collapsed to 0.144, indicating that the three competing loss signals created conflicting gradients that prevented effective learning of any individual task. Combined with bone features that lack positional information, this configuration represented the worst combination of excessive task complexity and insufficiently informative inputs.

4.8 Full System Comparison

Table 4.7 consolidates all experimental variants into a unified comparison, spanning the full pipeline from pose estimation through fall detection.

Four key findings emerge from the consolidated results. First, **the pose estimation head is the single most impactful component**. Replacing the 2D Heatmap with SimDR improved Keypoint AP from 0.573 to 0.802 (+40%), establishing the foundation upon which all downstream tasks depend. This gain propagated directly to the quality of keypoint features available for temporal action classification.

Second, **raw keypoints consistently outperformed bone features**. Transitioning from bone skeleton vectors (36 dimensions) to raw keypoint features (35 dimensions) improved Fall F1 from 0.680 (Skeleton-GRU) to 0.748 (Binary-BCE), confirming that absolute positional information—specifically the vertical location of the head relative to the bounding box—is the most discriminative cue for fall detection in thermal imagery.

Third, **the proposed dual-task method achieved the highest precision** (1.000) among all methods, producing zero false positives on the validation set. This characteristic makes it the most clinically safe option: when the system triggers a fall alert, care staff can fully trust it. The lower recall (0.502) compared to the Binary-BCE baseline (0.629) reflects the inherent challenge of jointly optimizing posture classification and fall detection with limited training data. The Binary-BCE baseline achieved the highest Fall F1 (0.748) by concentrating all model capacity on a single task with a simple loss—a well-established advantage of parsimonious models on small datasets.

Fourth, **synthetic data augmentation** is the anticipated pathway to unlocking the proposed architecture's full potential. The dual-task model possesses higher representational capacity that is currently underutilized due to data scarcity. With synthetically augmented training data, the proposed method achieves a Fall F1 of 0.776 (vs. 0.669 without augmentation, +16% relative) and recall of 0.655 (vs. 0.502), surpassing the Binary-BCE baseline (F1: 0.748), validating the architectural choice.

4.9 Qualitative Results

The system correctly handled the majority of common indoor monitoring scenarios, with fall detection typically triggering within 2–3 frames (~0.2–0.3 seconds at 10 FPS) of the fall onset. For standard walking and standing activities, the model consistently produced near-zero fall probabilities, demonstrating effective discrimination between routine activities and genuine falls. The proposed method's perfect precision meant that in practice, no false alarms were generated—when the system triggered an alert, clinical staff could trust the notification.

The primary failure mode involved slow, controlled falls that kinematically resembled deliberate sitting or lying-down transitions. In these cases, the keypoint trajectories lacked the sudden acceleration that characterizes uncontrolled falls, causing the model to under-score the event. Occlusion represented a secondary failure mode: when a person's limbs were partially occluded by furniture, the pose head occasionally produced inaccurate keypoint predictions, which propagated errors to the temporal action classifier. Multi-person tracking generally functioned well in low-occupancy scenes (1–3 persons), but tracking accuracy degraded under heavy spatial overlap, particularly when multiple individuals occupied adjacent thermal signatures with indistinguishable appearance.

4.10 Computational Cost

Table 4.8 reports the model size and inference latency of each component, measured on a single NVIDIA TITAN RTX GPU.

Table 4.8: Model size and inference speed (NVIDIA TITAN RTX).

Component	Parameters	Latency (ms)
Backbone (MobileViT-S)	4,937,632	6.9
Neck (CSPNeXtPAFPN)	2,912,080	3.6
Detection Head (RTMDet)	670,281	0.5
Pose Head (SimDR)	789,020	0.6
Action Head (GRU)	31,617	0.8
Backbone + Neck	—	10.7
Full Model	9,340,630 (~9.3M)	~12

1 The complete model contained only 9.3 million parameters, with the MobileViT-S backbone accounting for the largest share (4.9M, 53%). The temporal action head added negligible overhead: 31,617 parameters (0.3% of the total) and 0.8 ms latency per frame. The full pipeline achieved approximately 80 FPS on the NVIDIA TITAN RTX (backbone + neck: 10.7 ms, action head: 0.8 ms), well above real-time requirements. ByteTrack added less than 0.5 ms per frame. The lightweight design—9.3M parameters total—makes the system suitable for deployment on edge devices equipped with FPGA acceleration in elderly care facilities, aligning with the SensIR project’s target of >20 FPS on embedded hardware.

4.11 Discussion

The experimental results yield several insights that inform both the design of thermal infrared fall detection systems and the broader practice of multi-task learning on small datasets.

Pose representation is the most impactful design choice. SimDR’s disentangled 1D classification approach achieved a Keypoint AP of 0.802, dramatically outperforming the 2D Heatmap (0.573) and Direct Regression (0.570) alternatives. This +40% relative improvement established the foundation for all downstream tasks, as keypoint feature quality directly determines the action classifier’s input signal quality. The KL divergence loss with soft Gaussian targets proved particularly well-suited for the small RoI sizes (48×48) typical of thermal infrared images, where 2D heatmap quantization errors are proportionally larger than in standard RGB pose estimation settings.

Dual-task learning provides the most architecturally complete system. While the simpler Binary-BCE baseline achieved the highest Fall F1 (0.748), the proposed dual-task method offered a more architecturally sound solution by: (a) providing clinically valuable posture classification output that single-task approaches entirely lack; (b) achieving perfect fall precision (1.000), guaranteeing zero false alarms; and (c) possessing higher representational capacity that will benefit from additional training data. The performance gap was a direct consequence of the limited dataset size (4,838 instances) rather than an architectural limitation—a conclusion supported by our preliminary synthetic data augmentation results.

Simple approaches excel on small data, but complex architectures scale better. The

consistent ranking across our experiments—Binary-BCE > Binary-Focal > Dual-task on Fall F1—reflects the well-known bias-variance trade-off. Models with fewer optimization objectives and simpler loss functions generalize more readily with limited training examples. However, this advantage is expected to diminish as training data grows, at which point the richer representations learned by the dual-task architecture should yield superior performance. Indeed, with synthetic data augmentation, the proposed dual-task method achieves a Fall F1 of 0.776, surpassing the Binary-BCE baseline (0.748) and confirming that the architecture’s higher representational capacity becomes advantageous with sufficient training data.

The system is practical for real-world deployment. At 9.3M parameters and approximately 80 FPS on a single GPU, the system meets the real-time requirements for clinical monitoring with substantial headroom. The two-stage training strategy further enables modular development: the Stage 1 detection-plus-pose checkpoint can be shared across different downstream tasks (e.g., activity recognition, gait analysis) without retraining. The privacy-preserving nature of thermal infrared imagery, combined with the system’s lightweight architecture and end-to-end design, positions it as a practical solution for autonomous fall detection in elderly care facilities.

Chapter 5 Summary & Outlook

This chapter concludes the thesis by summarizing the contributions of the proposed thermal infrared fall detection system (Section 5.1) and outlining directions for future research (Section 5.2).

5.1 Summary

Falls represent one of the most significant health hazards facing the elderly population worldwide. According to the World Health Organization, approximately 684,000 fatal falls occur globally each year, and adults aged 60 and older account for the largest share of these incidents. In residential care facilities, timely detection of falls can dramatically reduce the severity of outcomes—studies have shown that lying on the floor undetected for more than one hour is independently associated with serious injury and increased mortality, even when the initial fall itself is not severe. Automated fall detection systems based on computer vision offer a promising approach to continuous, non-intrusive monitoring. However, conventional RGB camera-based methods raise fundamental privacy concerns: residents, families, and regulatory bodies are often reluctant to accept continuous video surveillance that captures identifiable imagery of individuals in their private living spaces. Moreover, RGB systems are inherently unreliable under the low-light conditions common in residential rooms during nighttime—precisely the period when fall risk is elevated due to disorientation, medication effects, and reduced visibility. Thermal infrared (IR) imaging provides a compelling alternative modality. By capturing emitted body heat rather than reflected visible light, thermal cameras preserve occupant privacy (no identifiable facial features or clothing details are recorded) and operate robustly regardless of ambient illumination. The central technical challenge, however, is that thermal IR images—particularly from low-cost thermopile sensor arrays such as the Heimann HTPA series used in the SensIR project—exhibit substantially lower spatial resolution and thermal contrast compared to RGB, rendering fine-grained human pose estimation and temporally nuanced action recognition considerably more difficult.

To address these challenges, this thesis presented an end-to-end multi-task framework for real-time fall detection from thermal infrared imagery. The system is built upon the RTMDet [38] anchor-free detection architecture and integrates three tightly coupled tasks within a unified pipeline. In the first stage, a person detection head localizes human subjects with bounding box regression using Quality Focal Loss [27] and GIoU loss [48], while a pose estimation head simultaneously predicts seven anatomical keypoints (head, shoulder center, both hands, hip center, both feet) via SimDR [28]—a disentangled 1D coordinate classification method that decomposes the 2D localization problem into two independent 1D classification tasks along the x - and y -axes. In the second stage, a temporal action classification head built upon a Gated Recurrent Unit (GRU) [9] processes sequences of normalized keypoint features extracted from tracked individuals. The proposed dual-task design of this action head simultaneously performs four-class posture classification

(standing, walking, sitting, lying) and binary fall detection, sharing a common GRU temporal encoder so that posture understanding directly informs fall discrimination. The backbone employs MobileViT-S [41], a hybrid CNN-Transformer architecture with only 4.9M parameters that combines the local feature extraction strength of convolutional layers with the global contextual modeling capability of self-attention, paired with a CSPNeXtPAFPN [38] neck for multi-scale feature aggregation across strides {8, 16, 32}. RoI features for each detected person are extracted via RoIAlign [19] at a fixed output size of 48×48 pixels. The two-stage training strategy—where Stage 1 jointly optimizes detection and pose estimation, and Stage 2 freezes all Stage 1 parameters before training only the randomly initialized temporal action head—enables independent development, evaluation, and modular replacement of each component without retraining the entire pipeline. Multi-object tracking via ByteTrack [74] with Kalman filter motion prediction maintains consistent person identities across frames, enabling the GRU to process coherent per-person keypoint sequences of up to 30 frames.

Comprehensive experimental evaluation on the SensIR thermal IR dataset yielded four principal findings. First, the choice of pose estimation representation proved to be the single most impactful design decision in the entire pipeline. As reported in Table 4.2, SimDR achieved a COCO Keypoint AP of 0.802, outperforming the 2D Heatmap baseline (0.573) and the Direct Coordinate Regression approach (0.570) by a relative margin of approximately 40%. At the stricter AP@0.75 threshold, SimDR's advantage was even more pronounced (0.874 vs. 0.574 and 0.595). The disentangled 1D classification formulation with KL divergence loss against soft Gaussian targets was particularly effective for the small RoI sizes (48×48) characteristic of thermal IR images, where the quantization error inherent in 2D heatmap argmax decoding is proportionally large relative to the spatial resolution. This finding has broader implications: SimDR's advantages may extend to other low-resolution pose estimation tasks beyond thermal imaging.

Second, the ablation study on input feature representations (Table 4.6) demonstrated that raw keypoint features—encoding absolute position (x, y), visibility, and inter-frame displacement ($\Delta x, \Delta y$) in 35 dimensions—consistently outperformed more complex alternatives. Bone skeleton features (36 dimensions), which encode translation-invariant relative joint displacements, achieved a Fall F1 of only 0.680, while adding 96-dimensional RoI appearance features further degraded performance to 0.641 due to the low discriminative value of thermal IR appearance and the increased overfitting risk from the $3.7\times$ dimensionality expansion. The decisive advantage of raw keypoint features confirms that absolute spatial information—particularly the vertical position of the head relative to the bounding box—is the most discriminative cue for fall detection, a finding that is intuitive given that falls fundamentally involve a transition from upright to horizontal body configuration.

Third, the fall classification strategy comparison (Table 4.3) revealed a nuanced trade-off between architectural completeness and data efficiency. The proposed dual-task method achieved a Fall F1 of 0.669 with *perfect precision* (1.000)—producing zero false positives on the validation set—while simultaneously providing clinically valuable four-class posture classification output. The simpler Binary-BCE baseline achieved a higher overall Fall F1 of 0.748 (Table 4.5), driven by superior recall (0.629 vs. 0.502). This performance gap was

144 attributed to the dual-task model's higher representational capacity leading to underfitting on the limited training set of 4,838 instances: the model must simultaneously learn to discriminate four posture classes and detect falls, distributing its learning capacity across two objectives. The loss function ablation further confirmed that on small datasets, simpler optimization landscapes (plain BCE with class weighting) outperform more sophisticated alternatives (focal loss, hard negative mining), which represent architectural investments that pay off only with sufficient training data. With synthetic data augmentation, the proposed dual-task method improves to a Fall F1 of 0.776 (recall: 0.655, precision: 0.951), surpassing the Binary-BCE baseline and confirming that the dual-task architecture scales favorably with additional data.

Fourth, computational profiling (Table 4.8) confirmed that the complete system is highly efficient: 9.3M total parameters with the full inference pipeline achieving approximately 80 FPS on an NVIDIA TITAN RTX GPU. The temporal action head contributes negligible overhead—only 31,617 parameters (0.3% of the total model) and 0.8 ms per frame—demonstrating that the GRU-based skeleton classifier adds rich temporal reasoning at virtually no computational cost. ByteTrack multi-object tracking adds less than 0.5 ms per frame. These characteristics make the system well-suited for deployment on edge devices equipped with FPGA or NPU acceleration, aligning with the SensIR project's target of >20 FPS on embedded hardware.

2 Despite the Binary-BCE baseline achieving higher Fall F1 on the current dataset, the dual-task architecture was deliberately selected as the final system because it provides the most architecturally complete pipeline for real-world clinical deployment. This decision was motivated by four considerations. First, the four-class posture classification output is directly useful for clinical monitoring applications that extend beyond acute fall detection—for instance, identifying prolonged lying episodes that may indicate a post-fall inability to self-recover, tracking declining daily mobility as an early warning of deteriorating health, or monitoring sleep patterns in residential care. Second, the perfect fall precision (1.000) guarantees zero false alarms, a property that is critical for clinical trust and adoption. Healthcare staff subject to frequent false alerts develop alarm fatigue, eventually disregarding or delaying response to notifications [52]; a system with zero false positives avoids this entirely. Third, the shared GRU encoder that jointly learns posture representations and fall discrimination forms a closed-loop design: understanding that a person is in a lying posture directly constrains the fall detection decision, and distinguishing between controlled lying-down transitions and uncontrolled falls requires precisely the posture-level understanding that the auxiliary task provides. Fourth, the architecture's higher representational capacity scales favorably with additional training data, as confirmed by the synthetic data augmentation results: the proposed method achieves Fall F1 of 0.776 with synthetic augmentation, surpassing both its baseline (0.669) and the Binary-BCE alternative (0.748).

108 In conclusion, this thesis demonstrates that thermal infrared fall detection can achieve real-time, privacy-preserving performance through a lightweight multi-task architecture that jointly addresses person detection, pose estimation, and temporal action classification within a single framework. The combination of SimDR-based pose estimation—which dramatically outperforms conventional heatmap and regression alternatives on low-

resolution thermal features—with a dual-task GRU temporal classifier provides both quantitative accuracy and clinical interpretability. The system's compact model size of 9.3M parameters, modular two-stage training pipeline, and approximately 80 FPS throughput make it a practical foundation for autonomous fall detection in residential and clinical care facilities, where privacy preservation, real-time responsiveness, and deployment on resource-constrained edge hardware are essential requirements.

5.2 Outlook

While the proposed system establishes a functional end-to-end pipeline for thermal IR fall detection and demonstrates the viability of the multi-task approach, several directions for future research can address its current limitations and substantially extend its capabilities. This section discusses five such directions, ordered roughly by their expected impact and feasibility.

5.2.1 Synthetic Data Augmentation and Dataset Expansion

The most pressing limitation of the current system is the modest size of the training dataset. The SensIR dataset contains only 4,838 training instances with 1,482 falling examples—a volume that is insufficient to fully exploit the dual-task model's representational capacity and that contributes directly to the performance gap relative to simpler single-task baselines. As demonstrated by the loss function ablation (Table 4.5), increasing model complexity or adding sophisticated loss components (focal loss, hard negative mining) yielded diminishing or even negative returns on this data volume, indicating that the primary bottleneck is data quantity rather than architectural expressiveness.

Two complementary strategies for synthetic data generation will be investigated. The first involves physics-based simulation of thermal IR sequences using parametric human body models (e.g., SMPL [34]) rendered under simulated thermal emission profiles. By varying body proportions, clothing insulation properties, ambient temperature, and camera viewpoint, a diverse corpus of synthetic falling and non-falling sequences can be generated with automatic ground-truth annotations for bounding boxes, keypoints, and action labels. The second strategy leverages domain adaptation from existing large-scale RGB fall detection datasets—such as the UR Fall Detection dataset [24], the Le2i Fall Detection dataset [5], or the NTU RGB+D dataset [53]—by learning an image-to-image translation mapping from RGB to thermal IR appearance using generative adversarial networks (GANs) or diffusion models, while preserving the original pose and action annotations. This approach benefits from the substantially larger volume and diversity of existing RGB fall datasets without requiring additional physical data collection.

The proposed dual-task architecture is expected to benefit disproportionately from additional training data compared to simpler baselines. The Binary-BCE model, with its single optimization objective and minimal architectural complexity, likely approaches its performance ceiling on the current dataset; additional data would provide diminishing returns. In contrast, the dual-task model's shared GRU encoder, which must learn both posture discrimination and fall detection simultaneously, requires a richer and more

diverse training signal to fully exploit the representational synergies between these two tasks. Preliminary experiments with synthetic data augmentation confirm this hypothesis: the proposed method achieves a Fall F1 of 0.776 with synthetic augmentation, surpassing both the base proposed result (0.669) and the Binary-BCE baseline (0.748).

5.2.2 Advanced Temporal Modeling

The current temporal action head employs a single-layer GRU processing fixed-length keypoint sequences of $T = 8$ frames with a per-frame MLP encoder that independently projects each frame's 35-dimensional keypoint vector into a 64-dimensional latent space before temporal aggregation. While this design is computationally efficient and sufficient for the current dataset, it has several inherent limitations that more expressive temporal architectures could address.

First, the fixed sequence length of $T = 8$ frames constrains the temporal receptive field to approximately 0.8 seconds at 10 FPS. While this window captures the acute phase of most falls, it is insufficient for distinguishing slow, controlled transitions (e.g., gradually sitting down over 3–5 seconds) from rapid falls that unfold over a similar duration but with different acceleration profiles. A Transformer-based temporal encoder [64] with multi-head self-attention and sinusoidal positional encoding would naturally accommodate variable-length input sequences and enable the model to attend selectively to the most discriminative frames within arbitrarily long temporal windows. The attention weight distribution would also provide interpretability—revealing which temporal moments the model considers most diagnostic of a fall.

Second, the per-frame MLP encoder treats the seven keypoints as a flat feature vector, discarding the spatial topology of the human body. Spatial-Temporal Graph Convolutional Networks (ST-GCN) [72] and their successors [54, 8] operate directly on the skeleton graph structure, applying graph convolutions that aggregate information along anatomical connections (e.g., head→shoulder→hip→foot). For our seven-keypoint skeleton, the graph structure is admittedly simple, but explicit graph reasoning could still improve the model's ability to detect coordinated multi-joint patterns characteristic of falls—such as simultaneous head drop and leg buckling—that a flat MLP might fail to capture. A hybrid GCN-Transformer architecture, combining spatial graph convolutions with temporal self-attention, represents a particularly promising direction.

Third, temporal multi-scale fusion could combine short-term motion cues (2–3 frames, capturing sudden acceleration and jerk) with medium-term context (8–15 frames, capturing postural transition dynamics) and longer-term behavioral patterns (30+ frames, capturing pre-fall gait instability). A multi-branch architecture with dilated temporal convolutions at different scales, fused via an attention-weighted aggregation, would provide the model with a richer temporal representation than the current single-scale GRU. However, all of these more expressive architectures increase model capacity and thus require substantially more training data to avoid overfitting, further motivating the synthetic data augmentation efforts described in Section 5.2.1.

5.2.3 Multi-Camera and Cross-Scene Generalization

The current model was trained and evaluated exclusively on data from a single Heimann HTPA thermopile sensor array installed at a fixed viewpoint in a controlled indoor environment. While this setup mirrors the intended deployment scenario—a ceiling-mounted sensor in a single room—the resulting model’s generalizability to different sensors, viewpoints, and environments remains an open question.

Domain adaptation across different thermal camera models presents a particularly important challenge. Thermal cameras vary widely in spatial resolution (from 32×32 thermopile arrays to 640×480 microbolometer arrays), noise equivalent temperature difference (NETD), dynamic range, and spectral sensitivity. A model trained on one camera type may fail catastrophically when deployed on another due to these systematic domain shifts. Unsupervised domain adaptation techniques—such as adversarial feature alignment or optimal transport-based distribution matching—that align the feature distributions of source and target camera domains without requiring labeled data from the target domain will be investigated. Camera-agnostic feature normalization, which removes camera-specific biases in the backbone’s intermediate representations, may provide a lightweight alternative.

Multi-view fusion from two or more cameras covering the same physical space opens the possibility of approximate 3D pose reconstruction from 2D thermal inputs. The SensIR project hardware already incorporates two thermopile sensors connected to a single edge device; fusing their overlapping fields of view through stereo calibration and triangulation could yield 3D keypoint coordinates, providing more robust fall detection by reasoning about body orientation relative to the gravity vector rather than relying solely on 2D projections that are inherently ambiguous under viewpoint changes.

Cross-environment generalization—encompassing diverse room layouts, thermal source configurations (radiators, sunlit windows, cooking appliances), furniture arrangements, and population demographics (varying body types, clothing insulation, mobility levels)—will require either collecting labeled data from a broad range of settings or developing self-supervised pre-training strategies. Contrastive learning on large unlabeled thermal video datasets, where the model learns to distinguish temporal neighbors from distant frames and spatial augmentations of the same scene from different scenes, could provide a robust initialization that transfers across environments without extensive per-site fine-tuning.

5.2.4 Clinical Integration and Edge Deployment

Transitioning from offline evaluation on recorded sequences to real-time clinical deployment introduces additional engineering, regulatory, and human-factors challenges that extend beyond pure algorithmic performance.

At the system integration level, the fall detection pipeline must be embedded within a hospital or care facility alert infrastructure. This requires designing a real-time notification workflow that combines instantaneous fall probability with temporal context to minimize nuisance alerts. For example, a clinically meaningful alert policy might require that (i) the

model's fall probability exceeds a threshold for at least two consecutive frames (eliminating single-frame transients), (ii) the detected person remains in a lying posture for more than 30 seconds without significant movement (distinguishing a fall from a deliberate lie-down), and (iii) the alert is escalated through a tiered notification system (bedside indicator → nurse station alert → emergency page) with configurable delays at each tier. The proposed system's dual-task output—providing both a binary fall score and a four-class posture state—is uniquely suited to this type of composite alerting logic, since posture duration monitoring requires explicit posture classification that single-task approaches do not provide.

On-device deployment targeting the FPGA acceleration hardware specified in the SensIR project, or alternatively NVIDIA Jetson Orin edge GPUs, will necessitate model optimization beyond the current PyTorch implementation. INT8 quantization of the MobileViT backbone and CSPNeXtPAFPN neck, with calibration on a representative thermal IR dataset, is expected to reduce inference latency by approximately 2× with minimal accuracy degradation. Operator fusion, batch normalization folding, and deployment via TensorRT or ONNX Runtime would further reduce overhead. The system's compact 9.3M-parameter footprint and the MobileViT backbone's demonstrated efficiency on mobile hardware [41] position it well for such optimization. The target throughput of >20 FPS on the SensIR FPGA platform appears achievable given the current 80 FPS performance on a desktop GPU.

Beyond acute fall detection, the four-class posture output enables long-term monitoring analytics that are clinically valuable in their own right. By aggregating posture classifications over days and weeks, the system could track daily mobility patterns (time spent standing vs. sitting vs. lying), sleep regularity (lying onset and wake times), and gradual gait deterioration (decreasing walking episodes or increasing sitting duration) as early predictive indicators of elevated fall risk. These longitudinal trends, presented through a clinical dashboard, could enable proactive interventions—such as physical therapy referrals or environmental modifications—before a fall occurs. Finally, a user study with clinical nursing staff will be essential to evaluate the system's practical utility: measuring alert response times, trust calibration (do staff act on alerts appropriately?), false alarm tolerance thresholds, and the value of posture classification information in daily care workflows.

5.2.5 Model Architecture Improvements

Several architectural refinements could improve accuracy and robustness as more training data becomes available. First, the hand-crafted per-keypoint importance weights could be replaced by a learned attention module that dynamically weights keypoint contributions based on temporal context, adapting to occlusion and different fall types. Physics-informed features—such as center-of-mass trajectory and head jerk—could provide stronger biomechanical inductive biases for distinguishing controlled movements from genuine loss-of-balance events.

Second, the current two-stage training introduces an information bottleneck: the pose head is optimized without knowledge of the downstream action task. **End-to-end joint training**

of detection, pose estimation, and temporal classification in a single loop could improve gradient flow and representation sharing, with uncertainty-based loss weighting to balance competing objectives. Finally, extending the system to model spatial relationships between persons and furniture—via a lightweight scene graph module—would improve robustness in multi-person and cluttered environments common in clinical care settings.

Appendix

Chapter A Source Code

A.1 Multi-Task Training Loss (RTMDetWithPose)

The loss method orchestrates the three training objectives: bounding-box detection, SimDR keypoint estimation, and dual-task action classification. All three heads share the same FPN feature pyramid; the pose and action heads additionally use RoI-aligned crops.

Listing A.1: RTMDetWithPose.loss – three-task combined loss

```

1 def loss(self, batch_inputs, batch_data_samples):
2     x = self.extract_feat(batch_inputs) # shared FPN features [P3,P4,P5]
3     losses = dict()
4
5     # 1. Detection: bbox + classification losses
6     det_losses, det_results = self.bbox_head.loss_and_predict(
7         x, batch_data_samples, proposal_cfg=self.pose_det_cfg)
8     losses.update(det_losses)
9
10    # 2. Pose: RoI-align person crops, then SimDR keypoint loss
11    rois = (self._build_gt_rois(batch_data_samples) # GT boxes + jitter
12            if self.pose_use_gt_box
13            else self._build_pose_rois(det_results, batch_data_samples)[0])
14    if rois.numel() > 0:
15        roi_feats = self.pose_roi_extractor(x, rois) # (N, C, 48, 48)
16        losses.update(self.pose_head.loss(roi_feats, rois,
17                                         batch_data_samples))
18
19    # 3. Action: temporal keypoint sequence -> dual-task GRU loss
20    if self.action_head is not None and rois.numel() > 0:
21        kpt_seq, gt_action = self._extract_temporal_action_data(
22            rois, batch_data_samples) # (N, T, 35), (N,)
23        if kpt_seq is not None:
24            losses.update(self.action_head.loss(kpt_seq, gt_action))
25
26    return losses

```

A.2 SimDR Pose Head

Forward Pass

Axis-wise average pooling decomposes the shared $128 \times 48 \times 48$ feature map into two 1-D sequences, each processed by an independent Conv1d head to produce per-keypoint probability distributions over 96 bins ($48 \times \text{scale} = 2$).

Listing A.2: SimDR axis-pooling

```

1 def forward(self, feats):
2     """Return (x_logits [N,K,96], y_logits [N,K,96], vis_logits [N,K])."""

```

```

3     shared = self._extract_shared(feats)    # (N, 128, 48, 48) shared Conv
      stack
4
5     x_logits = self.x_head(shared.mean(dim=2))  # pool H -> (N,128,W) -> (N,
      K,96)
6     y_logits = self.y_head(shared.mean(dim=3))  # pool W -> (N,128,H) -> (N,
      K,96)
7
8     vis_logits = self.vis_fc(self.vis_gap(shared).flatten(1))  # (N, K)
9     return x_logits, y_logits, vis_logits

```

Loss

Listing A.3: KL divergence loss and 1-D Gaussian target

```

1 def _loss_roi_v6(self, roi_feats, rois, batch_data_samples):
2     x_logits, y_logits, vis_logits = self.forward(roi_feats)
3     N, K, Lx = x_logits.shape; Ly = y_logits.shape[-1]
4
5     gt_x_dist, gt_y_dist, loc_weights, vis_labels = self._build_targets_v6(
6         rois, batch_data_samples, Lx, Ly, x_logits.dtype, x_logits.device)
7
8     # KL(target || pred) summed over bins, weighted by keypoint visibility
9     kl_x = F.kl_div(F.log_softmax(x_logits, dim=-1),
10                    gt_x_dist, reduction='none').sum(-1)  # (N, K)
11     kl_y = F.kl_div(F.log_softmax(y_logits, dim=-1),
12                    gt_y_dist, reduction='none').sum(-1)
13     loss_simdr = ((kl_x + kl_y) * loc_weights).sum(1).div(K).mean()
14
15     loss_vis = F.binary_cross_entropy_with_logits(
16         vis_logits, vis_labels, reduction='mean')
17
18     return {"loss_keypoint": self.simdr_loss_weight * loss_simdr
19           + self.vis_loss_weight * loss_vis}
20
21 def _gaussian_1d(self, length, center, device, dtype):
22     """Normalised 1-D Gaussian used as soft classification target."""
23     x = torch.arange(length, device=device, dtype=dtype)
24     g = torch.exp(-(x - center) ** 2 / (2 * self.sigma_1d ** 2))
25     return g / (g.sum() + 1e-8)

```

A.3 Dual-Task GRU Action Head (ActionHead)

Shared Temporal Encoder

Per-keypoint importance weights scale each joint's features before the encoder. A temporal residual connection $\mathbf{o} = \mathbf{h}_T + \text{GRU}_T$ stabilises training when the GRU is freshly initialised.

Listing A.4: ActionHead MLP + GRU + temporal residual

```

1 def _encode(self, kpt_features):
2     """(N, T, 35) -> (N, 64)"""
3     N, T, D = kpt_features.shape

```



```

4
5     # Per-keypoint importance scaling: [1.5, 1.0, 0.5, 0.5, 1.5, 1.2, 1.2]
6     if self._kpt_weight is not None:
7         K, kd = self.num_keypoints, self.kpt_dim
8         kpt_part = (kpt_features[:, :, :K*kd].view(N, T, K, kd)
9                     * self._kpt_weight.view(1, 1, K, 1))
10        kpt_features = kpt_part.view(N, T, K * kd)
11
12    # Per-frame MLP: Linear(35->64) + ReLU + Dropout + Linear(64->64) + ReLU
13    encoded = self.encoder(kpt_features.reshape(N * T, D)).reshape(N, T, -1)
14
15    # Temporal GRU + LayerNorm
16    _, h_n = self.gru(encoded)
17    gru_out = self.gru_norm(h_n[-1])          # (N, 64)
18
19    # Residual: last-frame encoder + GRU temporal context
20    return encoded[:, -1, :] + gru_out        # (N, 64)

```

Dual-Task Loss

Listing A.5: ActionHead.loss – 4-class CE and focal BCE with hard-negative weighting

```

1  # Label mapping: original 10-class -> 4-class activity and binary fall
2  ACTION_TO_ACTION4 = [0, 1, 2, 0, 3, 0, 3, 3, 3, 3] # stand/walk/sit/lie
3  ACTION_TO_FALL    = [0, 0, 0, 0, 0, 0, 1, 1, 1, 1] # classes 6-9 are falls
4
5  def loss(self, kpt_features, gt_labels):
6      """gt_labels: (N,) original 10-class labels, -1 = invalid."""
7      action_logits, fall_logits = self.forward(kpt_features)
8      valid = gt_labels >= 0
9      gt_valid = gt_labels[valid].long()
10
11     # Map 10-class labels to 4-class activity and binary fall targets
12     gt_action4 = self._action_to_action4[gt_valid]
13     gt_fall = self._action_to_fall[gt_valid]
14
15     # 1. Activity head: 4-class cross-entropy (label-smoothed)
16     loss_action = F.cross_entropy(action_logits[valid], gt_action4,
17                                  weight=self.class_weight)
18
19     # 2. Fall head: focal BCE
20     # pos_weight=2 for falls; hard_neg_weight=3 for lying-down/getting-up
21     fv = fall_logits.squeeze(-1)[valid]
22     p = torch.sigmoid(fv)
23     ce = F.binary_cross_entropy_with_logits(fv, gt_fall, reduction='none')
24     pt = p * gt_fall + (1 - p) * (1 - gt_fall)
25     focal = ((1 - pt) ** self.focal_gamma) * ce
26
27     sample_w = torch.ones_like(gt_fall)
28     sample_w[gt_fall == 1] = self.pos_weight[0]          # positive weight
29     for c in (self._hard_neg_classes or []):
30         sample_w[gt_valid == c] = self.hard_neg_weight   # hard-negative
31         weight
32     loss_fall = (focal * sample_w).mean()

```

```
33     return {'loss_action': loss_action * self.loss_weight,  
34           'loss_fall':    loss_fall    * self.fall_loss_weight}
```

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